

ASYMMETRIC J-CURVE IN NEPAL'S PEGGED EXCHANGE RATE REGIME

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By

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DECLARATION

I, Satish Chaudhary, hereby declare that this thesis entitled *ASYMMETRIC J-CURVE IN NEPAL'S PEGGED EXCHANGE RATE REGIME*, submitted to the Department of Economics, Ratna Rajyalaxmi Campus, is my own original work, except where other sources have been duly indicated and acknowledged. This thesis has not been submitted, in whole or in part, for any other degree or diploma at Tribhuvan University or any other institution. All sources of information used in this work have been acknowledged through proper citations.

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LETTER OF RECOMMENDATION

This thesis entitled *ASYMMETRIC J-CURVE IN NEPAL'S PEGGED EXCHANGE RATE REGIME* has been prepared by Mr. Satish Chaudhary under my guidance and supervision. I hereby recommend this thesis for the examination by the Thesis Committee as a partial fulfillment of the requirements for the Degree of MASTER OF ARTS in ECONOMICS.

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LETTER OF APPROVAL

We certify that this thesis entitled “*ASYMMETRIC J-CURVE IN NEPAL'S PEGGED EXCHANGE RATE REGIME*” submitted by Mr. Satish Chaudhary to the Central Department of Economics, Faculty of Humanities and Social Sciences, Tribhuvan University, in partial fulfillment of the requirements for the Degree of MASTER OF ARTS in ECONOMICS has been found satisfactory in scope and quality. Therefore, we accept this thesis as a part of the said degree.

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ABSTRACT

The study looks at whether an asymmetric J-Curve exists in Nepal's trade balance under its dual exchange rate regime: a fixed peg with the Rupee and a floating arrangement against the Rest of the World (ROW). Nepal's trade deficit has widened even though the currency keeps depreciating and studies that used linear models found no J-Curve. Instead those studies suggested L-Curve." Using data from 1980 to 2024 this study applies Linear ARDL bounds testing, Error Correction Modelling and the Non-linear ARDL (NARDL) framework one after the other to look at the overall trade balance, bilateral trade with India and trade with the ROW. Cointegration and a significant long-run asymmetry appear in the ROW regime, where real depreciation makes the trade balance worse while real appreciation does not have a significant effect. Trade with India shows no cointegration and a symmetric adjustment, which suggests that the bilateral peg stops exchange rate movement from affecting trade. Therefore the J-Curve hypothesis is rejected. Nepal's trade with the ROW follows a "Worsening Curve" in which depreciation deepens the deficit over time which's in line, with the failure of the Marshall-Lerner condition. The study concludes that exchange rate policy alone cannot fix Nepal's trade deficit and recommends reforms, export diversification, inflation control and the productive use of remittances.

Keywords: J-Curve, asymmetric cointegration, trade balance, Real Effective Exchange Rate

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ABBREVIATIONS

ADF	Augmented Dickey-Fuller (test)
AIC	Akaike Information Criterion
ARDL	Autoregressive Distributed Lag
BG	Breusch-Godfrey (test)
BPG	Breusch-Pagan-Godfrey (test)
COVID-19	Coronavirus Disease 2019
CUSUM	Cumulative Sum
CUSUMSQ	Cumulative Sum of Squares
ECM	Error Correction Model
ECT	Error Correction Term
Exchange_rate_npr	Nominal exchange rate of NPR against USD
Exp_India_M_NPR	Exports to India (NPR, millions)
Exp_ROW_M_NPR	Exports to Rest of the World (NPR, millions)
Exports_M_NPR	Total exports (NPR, millions)
GDP	Gross Domestic Product
GDP_USD	Gross Domestic Product (USD)
Imp_India_M_NPR	Imports from India (NPR, millions)
Imp_ROW_M_NPR	Imports from Rest of the World (NPR, millions)
Imports_M_NPR	Total imports (NPR, millions)
INR	Indian Rupee
JB	Jarque-Bera (test)
LDCs	Less Developed Countries

ln_india_gdp	Natural log of India's real GDP
ln_india_tbr	Natural log of bilateral Trade Balance Ratio with India
ln_nepal_gdp	Natural log of Nepal's real GDP
ln_reer	Natural log of the Real Effective Exchange Rate
ln_row_gdp	Natural log of Rest of the World real GDP
ln_row_tbr	Natural log of bilateral Trade Balance Ratio with ROW
ln_total_tbr	Natural log of aggregate Trade Balance Ratio
ML	Marshall-Lerner (condition)
NARDL	Non-linear Autoregressive Distributed Lag
NEER	Nominal Effective Exchange Rate
NPR	Nepalese Rupee
NRB	Nepal Rastra Bank
PP	Phillips-Perron (test)
REER	Real Effective Exchange Rate
REER ⁻ / REER_NEG	Negative partial sum of REER (cumulative real depreciation)
REER ⁺ / REER_POS	Positive partial sum of REER (cumulative real appreciation)
RESET	Regression Equation Specification Error Test
ROW	Rest of the World
SIC	Schwarz Information Criterion
TB	Trade Balance
TBR	Trade Balance Ratio (exports divided by imports)
TBR_India	Bilateral Trade Balance Ratio with India
TBR_ROW	Bilateral Trade Balance Ratio with Rest of the World

TBR_total	Aggregate Trade Balance Ratio
TU	Tribhuvan University
USD	United States Dollar
VAR	Vector Autoregression
WDI	World Development Indicators

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Economists have debated the relationship between exchange rate fluctuations and trade flows for decades and the answer remains unclear. Many nations Nepal included, turned to currency devaluation to tackle trade deficits after Bretton Woods collapsed in the 1970s. The theory says that if a currency becomes cheaper exports should look more attractive, to buyers while imports become more expensive for domestic consumers and the trade balance should slowly improve. However the real world economy rarely works in this way.

In the term the amount of trade often stays the same because of things that don't change easily. These things include contracts, delays in shipping and people who like to buy the same things they always have. This situation, where a drop in the value of a currency makes the trade deficit worse at first before it gets better is called the "J-Curve effect". In the past some studies did not find this effect. Their results were often because they thought the economy would react the way to a 5% rise in value as it would to a 5% fall. More recent researchers have questioned this idea. Shin et al. (2014) created a model called the Non-linear Autoregressive Distributed Lag (NARDL) model. This model helps study how currency changes affect trade in ways. With this method Nasir and Leung (2021) showed that when the US dollar gets weaker it usually helps the trade balance. When the dollar gets stronger it hurts the trade balance in a way. Also, Arize et al. (2017) found that in a group of eight countries trade flows changed in a -linear way most of the time.

In the Asian context the evidence is more ambiguous. Bhat and Bhat (2021) found that while India does not show a J-Curve depreciation has a stronger positive effect, on the trade balance than appreciation has a negative effect. Ganai and Khan (2025) made the picture more complicated by saying that even though depreciation gives long-term benefits, industrial capacity, not currency value drives lasting surpluses. Together these studies show that exchange rate policy cannot be seen as a solution.

There is also growing evidence that aggregate data often hides the patterns inside. Bahmani-Oskooee and Harvey (2017) showed that linear models used on Singapore's trade could not see relationships that only appeared when looking at bilateral trade. Bahmani-Oskooee et al. (2019) repeated this idea when they broke US-China trade into 97 product categories and found industry-level J-Curves that were not visible, in the totals. Likewise Bahmani-Oskooee and Karamelikli (2018) discovered that currency changes affect Japan-US trade in ways depending on the mix of commodities.

Nepal stands out in this body of research. The country keeps a fixed exchange rate with the Rupee meaning the Nepalese Rupee is always tied to the Indian Rupee. Because of this the real effective exchange rate of the Nepalese Rupee fluctuates when other currencies move against the Rupee or when inflation differs between Nepal and other countries. This leads to a multilateral exchange rate situation. What is most surprising is that Nepal goes against economic thinking. For the twenty years the Nepalese Rupee has been gradually losing value compared to many other convertible currencies. Normally a weaker currency should make exports cheaper. Imports more expensive, which would help reduce a trade deficit. In Nepal, the trade deficit has actually grown larger. This contradiction has stayed in place because previous studies used linear models. These models are too simple. They don't account for the messy and unpredictable nature of Nepal's trade behavior.

If depreciation is not fixing the deficit the problem probably comes from the way the effect is transmitted. Nepalese exporters may not have productive capacity to react to price changes. The economy might depend too much on essential imports such as fuel, food and machinery which makes demand inelastic. In that case depreciation only raises costs and does not lower volumes. The different behaviors of the fixed market and the floating global market may also cancel each other out when we look at overall data. To close this gap this study uses the Non-Linear Autoregressive Distributed Lag (NARDL) method to find out if the missing J-Curve in Nepal is due to a model or to deep structural problems, in the economy.

1.2 Statement of Problem

For years Nepal has faced a long-standing trade deficit. This trade deficit keeps getting larger. This growth goes against what normal economic theory would predict. Traditional theory says that when a currency falls in value it should help exports become cheaper and discourage imports.. Nepal shows a strange reality. Even though the Nepalese Rupee (NPR) has been falling against the US Dollar for the twenty years the trade deficit has not become smaller. Instead the trade deficit has grown a lot.

This discrepancy appears in the literature, where results still show large differences. Some scholars say that exchange rate changes will eventually fix the problem while others say that Nepal's trade elasticity is too small to make a difference. Much of this disagreement comes from limits. By using mostly linear models past research has assumed symmetry meaning the economy reacts the way when the currency rises or falls. Such a simplification probably hides the asymmetric dynamics of the Nepalese economy. Also by looking at the trade balance as one number these studies often miss that exports and imports may react to price changes in different ways and, at different times.

Nepal's monetary duality adds to the complexity of its situation. The country keeps a fixed exchange rate with the Indian rupee while allowing its currency to float against other global currencies. This dual system raises a question: does the trade balance respond differently under each regime? The central issue is why devaluation often fails to reduce the trade deficit. Is it because domestic production cannot keep up? Is it due to a deep reliance on essential imports? Could it also be that the models used to assess these effects are too simplistic? This study looks at the paradox by using a non-linear approach to directly explore the imbalances, between the two systems.

1.3 Research Questions

The following research questions guide this study:

1. What is the nature of the asymmetric J-Curve effect in Nepal's overall trade balance?
2. How does Nepal's trade balance react in a way when the real currency goes up in value compared to when it goes down in value over a long period of time ?

3. How are the asymmetric J-Curve effects different when looking at Nepal's trade with India, which has a fixed exchange rate and Nepal's trade, with the rest of the world, where the exchange rate is determined by the market?

1.4 Objectives of Research

The primary objective of this study is to evaluate the asymmetric and dynamic impact of real exchange rate fluctuations on Nepal's aggregate trade balance. To achieve this, the study pursues the following three specific objectives:

1. To examine the nature of the asymmetric J-Curve effect in Nepal's aggregate trade balance
2. To assess the asymmetric response of Nepal's trade balance to real currency appreciation versus real currency depreciation in the long run.
3. To analyze the asymmetric J-Curve effects differ between Nepal's trade with India (pegged regime) and the Rest of the World (floating regime) from 1980 to 2024.

1.5 Research Hypotheses

Hypothesis 1: Existence of the Asymmetric J-Curve effect in Aggregate Trade

This hypothesis tests whether the J-Curve pattern – initial short run deterioration of the trade following real depreciation, followed by long-run improvement – exists in Nepal's aggregate trade balance.

H0: There is no asymmetric J-Curve effect on Nepal's aggregate trade balance

H1: An asymmetric J-Curve effect exists in Nepal's trade balance.

Hypothesis 2: Asymmetric Adjustment to Appreciation versus Depreciation

This hypothesis tests whether real currency appreciation and depreciation exert statistically different effects on the trade balance in the long-run.

H0: The trade balance responds symmetrically to real appreciation and real depreciation in the long-run; the Wald test fails to reject $\theta^+ = \theta^-$ in the NARDL long-run coefficients.

H1: The trade balance responds asymmetrically in the long-run; the Wald test rejects $\theta^+ = \theta^-$ at the 5% significance level.

Hypothesis 3: Regime-Dependent Asymmetry

This hypothesis tests whether the asymmetric response of the trade balance to real exchange rate movements differs between the India (pegged) and ROW (floating) segments of Nepal's trade.

H0: The asymmetric response of the trade balance to real exchange rate movements is identical for India and ROW trade; in particular, the long-run Wald test of symmetry yields the same outcome in both regimes.

H1: The asymmetric response differs between the India and ROW regimes.

1.6 Limitations of Study

The limitations of the study are as follows:

First, this study uses annual data instead of other frequency data for the 45-year sample period from 1980 to 2024. Annual data give a historical view but they may smooth out very short-term fluctuations and mix the early deterioration and recovery phases of the J-Curve into a single observation. Therefore the short-run effects should be read as year-one impacts than adjustments that happen otherwise. This was a trade-off to keep all variables consistent.

Second, the sample period contains exogenous shocks, such as the 1990 trade liberalization the 2008-2009 global financial crisis, the 2015 Gorkha Earthquake and the COVID-19 pandemic. These events can create breaks that have non-linear effects on trade volumes. To reduce this risk the study uses CUSUM and CUSUMSQ tests to check parameter stability so that the estimated coefficients stay reliable after these disruptions.

Third, Nepal receives worker remittances about 25% of GDP which strongly influence the current account and import capacity. These inflows may interact with exchange rate dynamics by keeping import demand no matter how the currency moves potentially biasing the long-run exchange rate coefficients. Although remittance data are available adding them as an extra regressor is outside the scope of this study and points to a topic for research.

A further limitation concerns the exchange rate measure used in the model. The study uses the REER for both the India trade balance and the ROW trade balance. REER works well for trade and ROW trade but a bilateral real exchange rate (NPR/INR adjusted for inflation) would be more precise for the India trade balance. Unfortunately consistent data for such a series from 1980 to 2024 are not readily available. Future research could build this series.

Finally a methodological limitation comes from the lag structure chosen by the information criteria (AIC and SIC). These criteria selected lag-0 for both the negative partial sums in all three trade balance regimes. With this specification short-run asymmetry cannot be formally tested so the present study examines only long-run symmetry. This choice follows the Shin et al. (2014). Is justified by the limited degrees of freedom in the sample ($T \approx 43$) as well as by the fact that Nepal's de facto peg to the Indian rupee dampens short-run transmission. Future research, with a sample might look at short-run asymmetry once more annual observations are available.

1.7 Organization of Study

The study is divided into five chapters. The first chapter introduces the topic that includes the background of the study, the statement of the problem, the objectives, the formulation of the hypothesis, the limitations and the overall structure of the study. The second chapter reviews existing literature covering both empirical studies and highlights the research gap. The third chapter explains the research methodology, including the research design, the nature and scope of the data the tools and techniques used for data collection and the model specification. The fourth chapter focuses on the analysis and presentation of the data. The fifth and final chapter offers a summary the conclusions drawn from the research and the recommendations based on the findings.

CHAPTER II

REVIEW OF LITERATURE

The relationship between exchange rate fluctuations and the trade balance has been a contentious subject in international economics for decades. Specifically, the J-Curve phenomenon proposed by Magee (1973) suggests that currency devaluation or depreciation can worsen the trade deficit in the short term before it improves. Over the years, extensive empirical research has sought to establish this relationship, with mixed results. Some of the earliest studies found evidence supporting the J-Curve hypothesis (Bahmani-Oskooee, 1985). However, others failed to find any statistically significant relationship (Rose & Yellen, 1989). This quandary has lasting implications for macroeconomic policies, especially in developing countries like Nepal, where the trade deficit is a persistent problem exacerbated by the lack of policy instruments to address it. Therefore, the question whether the J-Curve applies in Nepal is of great practical and academic significance. It is crucial to determine whether currency devaluation can be an effective policy instrument to address the trade deficit.

This chapter presents a review of the existing literature on the J-Curve phenomenon and the theoretical background of the study. Furthermore, it explores the relevant empirical literature to identify the research gap that this study will address. It concludes by describing the theoretical and mathematical background of the Non-linear Autoregressive Distributed Lag (NARDL) approach, which this study will employ to explore the J-Curve paradigm in Nepal. The first section of this chapter provides a brief theoretical background of how trade and exchange rates interact. It starts with the Marshall-Lerner condition and demand for trade model proposed by Goldstein and Khan (1985). These two theories allow understanding the mathematical basis of the relationship between exchange rates and the trade balance. This section concludes by presenting the theoretical background of the J-Curve phenomenon. It starts by describing Magee's (1973) initial thoughts on the subject based on the currency-contract and pass-through framework. It then proceeds to explain the evolution of this theory to the degree it is understood today using the variables of the US trade balance, prices of imports and exports, and industrial production levels. The last

part of the theoretical background discusses the concept of asymmetric effects, which forms the basis of this study's approach.

The second section of this chapter is a literature review, which is split into two parts. The first part examines the works on the J-Curve phenomenon published in the 1980s and 1990s. It discusses the most notable contributions to the understanding of the J-Curve and identifies limitations of the early studies. The second part of the review focuses on the NARDL approach, which has been used in several studies exploring the J-Curve effect. The first study to use the NARDL methodology was Shin et al. (2014), who identified its applicability in the study of the J-Curve phenomenon. It was followed up by Bahmani-Oskooee and Fariditavana (2016), who discussed the mathematical background and the application of NARDL in the study of the J-Curve effect around the world. The third section applies the concepts from the first two sections to South Asia and Nepal in particular. It reviews the most relevant works, including the studies by Chaulagai (2015), Pathak (2020), Adhikari (2020), and Bishwakarma et al. (2024). The purpose of this section is to identify the research gap that this study will address. The final part of the chapter concludes by defining the theoretical and mathematical basis of this study's methodological framework and research question before proceeding to describe the objectives of this study.

2.1 Theoretical Foundations

The standard setup for modeling the interaction between exchange rates, income, and the trade balance is the imperfect substitute's model pioneered by Goldstein and Khan (1985). The model departs from the classical premise that goods traded across borders are perfect substitutes by assuming that domestically-produced and foreign goods are imperfect substitutes. The demand for a country's imports and exports is thus derived from two sources: the relative prices of the traded goods (real exchange rate) and the purchasing power of the country's residents (domestic and foreign income).

The model specifies the import demand function of a given country as $M = f(Y, P_m/P_d)$, where Y is the level of domestic income and P_m/P_d is the ratio of the prices of imports to domestically-produced goods. The same country's export demand function takes the form $X = f(Y, P_x/P_f)$, where Y is the level of foreign income and P_x/P_f is the ratio of the prices of exports to foreign-produced goods (Goldstein & Khan, 1985). Summing up

the import and export demand functions, it is evident that the trade balance is a function of the level of domestic and foreign income and the real exchange rate: $TB = f(Y, Y, REER)$ - the reduced-form version of the model applied in the empirical analysis of Rose and Yellen (1989), among others.

The specification of the trade balance as a function of domestic and foreign income informs the selection of variables in the econometric analysis. A regression model investigating the impact of exchange rates on the trade balance must control for the level of domestic and foreign income to isolate the effect of relative prices (Goldstein & Khan, 1985; Rose & Yellen, 1989). Failure to account for the income channel would result in the omitted variable bias as changes in the foreign income (Y) may be confounded with the changes in the exchange rate (Goldstein & Khan, 1985). The imperfect substitutes model thus justifies the inclusion of the real GDP of Nepal as a proxy for domestic income in the ARDL (Pesaran et al., 2001) and NARDL (Shin et al., 2014) specifications used in this study: the foreign income channel is accounted for at the stage of the pilot estimation and dropped in the final specification in favor of a lower degree of freedom following the warning of the consequences of its omission in the case of small-sample NARDL modeling (Abegaz, 2024).

While the imperfect substitute's model specifies the set of variables affecting the trade balance, the direction of the association is determined by the price elasticities of the demand for imports and exports (Goldstein & Khan, 1985; Rose & Yellen, 1989). This consideration gives rise to the Marshall-Lerner condition.

The analytical discussion of the transmission mechanism of exchange rate fluctuations to trade balance begins with the Marshall-Lerner (ML) condition. Although this condition was theoretically anticipated in Marshall's writings on money, credit, and commerce, the mathematical formulation of the condition is primarily attributed to Lerner (Goldstein & Khan, 1985; Rose & Yellen, 1989). Essentially, the Marshall-Lerner condition hypothesizes that devaluation of the domestic currency will improve the trade balance in the long run (with constant terms of trade) if the sum of the absolute values of the price elasticities of demand for exports and imports exceeds one (Goldstein & Khan, 1985; Rose & Yellen, 1989).

The implication of the ML condition is that when a domestic currency is devalued, its exports become cheap while its imports become expensive. While this hypothetical situation suggests an increase in the volumes of exports and a decrease in the volumes of imports, the price-volume effects of devaluation need further theoretical discussion. Specifically, while the Marshall-Lerner condition stipulates that devaluation will improve the trade balance, whether this outcome will be realized depends on the relative responsiveness of the volume of exports and imports to changes in prices (Rose & Yellen, 1989).

More precisely, if the demand for exports and imports is very responsive to changes in prices (i.e., if $\eta_X + \eta_M > 1$, where η_X and η_M indicate the price elasticity of export and import demand, respectively), the volume effect dominates the price effect, so that trade balance moves in the direction of improving the trade balance (Rose & Yellen, 1989). However, if demand is not very responsive to changes in prices (i.e., if $\eta_X + \eta_M < 1$), then devaluation worsens the trade balance because the price effect dominates (i.e., the price effect increases the cost of imports and reduces the price of exports).

One of the most influential contributions to the theoretical discussion of the Marshall-Lerner condition and its implications for the trade balance was made by Rose and Yellen (1989). The authors demonstrated that while the Marshall-Lerner condition is necessary for a short-term improvement of the trade balance, it is not sufficient. Specifically, authors showed that when the elasticity condition fails to hold, devaluation does not improve the trade balance. However, if the elasticity condition is satisfied, the trade balance will still decrease in the short term before increasing in the long term. In other words, authors provided the impetus for the theoretical discussion of the J-Curve phenomenon by demonstrating that devaluation can worsen the trade balance immediately after implementation before gradually improving it over time. This conclusion was reached as a result of examining the difference between the short- and long-term effects of devaluation on the trade balance.

The theoretical discussion of the Marshall-Lerner condition and related concepts is particularly relevant to countries such as Nepal, which may be considered developing, landlocked economies. As mentioned above, the theoretical model suggests that the demand for imports and exports will be highly responsive to changes in prices in elastic

markets and less responsive in inelastic markets. In the former case, the Marshall-Lerner condition holds, while in the latter, it does not. Inelastic demand for imports would imply that devaluation will lead to increased prices for imports without a significant change in their volumes, which will have a negative impact on the trade balance. Given the list of potential imports provided by Pathak (2020), it is possible to speculate on the likely elasticity or inelasticity of demand for them. As imports of petroleum, pharmaceuticals, and machinery are most likely inelastic, the Marshall-Lerner condition is unlikely to hold in Nepal. This conclusion is consistent with the findings of Pathak (2020) and provides the impetus for the discussion of the J-Curve phenomenon in the empirical analysis of Chaulagai (2015).

The J-Curve phenomenon was first put into shape by Magee (1973). Magee saw that the timing of trade contracts is key to how changes in exchange rates affect a country's trade balance. Magee's main idea was that there are "adjustment lags": because trade contracts are usually written in currencies and are revised only rarely the impact of a devaluation does not show up all at once. Import contracts that were signed before the devaluation still run at the rates so the price of imports rises immediately for the country. At the time the advantage of having cheaper exports does not show up right away because new contracts have to be negotiated at the lower new rates. This mismatch in timing creates the J-shaped path: first the trade balance gets worse than it gets better as the larger volume of trade starts to dominate the price changes.

Modern econometric work has turned this story into a statistical test. Bahmani-Oskooee (1985) created the empirical design that many later studies use. He used data from 1973 to 1980 for four developing countries—Greece, India, Korea and Thailand. He built one trade balance equation that included an Almon distributed lag (a third-degree polynomial with up to 12 lags) on the exchange rate. In that set-up the J-Curve appears if the trade balance gets worse after a devaluation and then gets better. Bahmani-Oskooee checked this by looking at whether the lag coefficients on the exchange rate (E/P) start positive and then become negative. The results showed a J-Curve for Greece, India and Korea. Not for Thailand. Today researchers look at the signs and importance of coefficients in Error Correction Models (ECM) or Non-linear Autoregressive Distributed Lag (NARDL) models (Shin et al. 2014; Bahmani-Oskooee & Fariditavana 2016). In those

models a J-Curve is confirmed when the short-run coefficient for a depreciation is negative and statistically significant while the long-run coefficient is positive and statistically significant. Rose and Yellen (1989) sharpened this approach by adding hypothesis tests to separate the short-run and long-run effects.

Let us look at Nepal to make this idea clear. Imagine the Nepalese Rupee (NPR) falls by 10% against the Rupee (INR). After the fall the price of goods that Nepal buys from India goes up by 10% in NPR terms. Because the existing contracts are "sticky" Nepalese importers have to pay more for the amount of goods and the trade deficit grows immediately. At the time Nepalese goods become cheaper for Indian buyers but the amount of exports does not jump right away. Indian companies need time to see the price drop and to renegotiate their purchase agreements. So the trade balance gets worse in the run (Magee, 1973). After some time—when buyers start buying more Nepalese goods and new contracts are signed—does the larger volume of trade begin to offset the higher prices? This could make the deficit smaller. Even turn into a surplus. The question of whether this adjustment happens between Nepal and India is exactly what Adhikari (2020) looks at.

The strength and length of this effect are not the same they depend on several conditions. First how "sticky" the contracts are decides how deep the initial drop will be. Countries that use longer-term rigid contracts usually see a J-Curve (Magee, 1973). Second the price elasticity of demand for exports and imports controls how quickly the trade balance recovers. The higher the elasticities, the move into the improving phase (Bahmani-Oskooee, 1985). Third the type of exchange rate system—floating, fixed or pegged—sets how fast the adjustment happens (Lal & Lowinger, 2002). Finally the mix of goods that a country trades and the nature of its trading partners matter a lot. Industry data show that J-Curve effects differ a lot between product categories (Bahmani-Oskooee & Hajilee, 2009). All of these factors explain why the J-Curve, even though it is a theory gives mixed results, in different countries and different times.

2.2 Beyond Symmetry: The Case for Asymmetric Effects

Traditional econometric frameworks that analyze the relationship between exchange rates and trade balances have long worked under the assumption of symmetry the idea that currency appreciation and currency depreciation apply opposite pressures on the trade balance. This assumption was built into linear models, which used a single

exchange rate coefficient to describe the relationship (Rose & Yellen, 1989; Bahmani-Oskooee, 1991). This orthodox approach has been increasingly challenged in recent literature leading to non-linear and asymmetric models that let different effects appear depending on the direction of the currency movement (Arize et al., 2017).

The theoretical justification for asymmetry comes from economic frictions. First import demand in developing economies is often very inelastic for essential goods. When the currency depreciates the prices of imports—such as pharmaceuticals, petroleum and industrial machinery—rise; yet because these goods are essential for survival and production import volumes do not fall as much. In contrast when the currency appreciates the drop in import prices may not cause a rise in volume because domestic demand for these essentials is already near its limit—a pattern that matches the elasticity estimates reported for Nepal (Pathak, 2020). Second the mix of trade is often different between exports and imports: developing nations usually export commodities that are more price-responsive while importing manufactured goods that are inelastic—industry evidence confirms that trade responses can vary sharply across product categories (Bahmani-Oskooee & Hajilee, 2009). Third market structures, in oligopolistic industries often produce "sticky" pricing, where firms are quicker to raise prices during depreciation to protect margins than to lower them during appreciation. Finally government actions—such as tariffs and quotas—add non-linearities that depend on whether the currency's rising or falling. Together these frictions predict the directional asymmetry that asymmetric cointegration studies have found for South Asian economies (Bhat & Bhat, 2021).

The empirical trigger for this shift was provided by Bahmani-Oskooee and Fariditavana (2015) whose research changed how J-Curve analysis is done. Their work showed that the symmetry assumption in linear models was too restrictive and often hid non-linear dynamics. By splitting exchange rate changes into appreciation) and negative (depreciation) parts they discovered that the effects on trade balances often differ not only in size but sometimes even in sign. Their findings suggested that linear models by forcing symmetry could mislead people about the existence and nature of the J-Curve—a risk that appeared in studies, where J-Curve effects invisible under linear ARDL became clear once asymmetry was allowed (Bahmani-Oskooee & Fariditavana, 2016; Iqbal et al., 2021). This

methodological insight is the reason for using the Non-Linear Autoregressive Distributed Lag (NARDL) framework in the present study.

The policy implications of asymmetry are especially sharp for Nepal. If exchange rates move asymmetrically policymakers cannot assume that a devaluation will mirror the effects of an appreciation. Because Nepal relies heavily on imports it is very likely that a currency depreciation will hurt the trade balance more than a currency appreciation will help it—a pattern already visible in the region, where India's trade balance reacts asymmetrically to currency changes (Bhat & Bhat, 2021) and where early Nepalese evidence points in the same direction (Adhikari, 2020). In such a case currency devaluation becomes a less useful tool, for fixing deficits than symmetric models would suggest (Chaulagai, 2015). Therefore adopting the NARDL method is essential to uncover these transmissions and to give a more accurate diagnosis of Nepal's trade dynamics.

2.3 Global Empirical Evidence

The empirical investigation of the J-Curve has produced a body of literature that spans almost forty years with many studies appearing in top journals of international economics. The first empirical studies laid out methods and showed the J-Curve phenomenon in several groups of countries (Bahmani-Oskooee, 1985; Gylfason & Risager, 1984). Later studies improved those methods. Widened the scope to cover developing countries, emerging markets and transition economies (Bahmani-Oskooee, 1991; Grigoryan, 2015; Kallianiotis, 2022). This section gives a review of the early empirical literature that set the research agenda for J-Curve studies focusing on results and new methods.

The study by Bahmani-Oskooee, (1985) was one of the most influential empirical investigations of the J-Curve phenomenon in Less Developed Countries. The research used data from 1973 to 1980 for four developing economies—Greece, India, Korea and Thailand. Bahmani-Oskooee (1985) estimated a trade balance equation that incorporated an Almon distributed lag, which is a third-degree polynomial with up to twelve lags on the exchange rate variable. The study tested whether the distributed lag coefficients on the exchange rate (E/P) began with a sign and then changed to a negative sign, which is the defining characteristic of the J-Curve. The results supported the J-Curve in three of the four

countries (Greece, India and Korea). Not in Thailand. Importantly the study highlighted the need to separate the short-run and long-run effects of devaluation. This distinction guided subsequent studies, including the error-correction frameworks that used separate short-run and long-run coefficients. Those frameworks became the empirical approach in the following decades (Rose & Yellen, 1989).

Bahmani-Oskooee (1991) extended this line of work by exploring the long-run relationship between the trade balance and the real effective exchange rate through cointegration techniques. The research used data from 1973 to 1988 and discovered that the trade balance and the real effective exchange rate of some Less Developed Countries are cointegrated over the long term. This finding was important because it brought cointegration into analysis. Cointegration allows researchers to test for long-run equilibrium relationships while still considering short-run changes. Bahmani-Oskooee (1991) argued that this method is a substitute for testing the Marshall-Lerner condition. The results supported a long-run equilibrium relationship in cases thereby helping to establish the cointegration framework that would become common in later J-Curve studies.

Rose and Yellen (1989) made a contribution by testing for the J-Curve with U.S. Aggregate data covering the previous twenty-five years. Their method tested whether the trade balance and the real exchange rate were cointegrated and whether the short-run and long-run coefficients showed the expected signs. The results showed no reliable evidence of a stable J-Curve in the U.S. Data. The study highlighted that the J-Curve is not a phenomenon and that the relationship between the exchange rate and the trade balance may be more complex than first assumed.

Gylfason and Risager (1984) investigated whether devaluation improves the account using a general macroeconomic model that connects devaluation foreign interest payments and the current account. The model included both demand-side effects (through exports, imports and spending) and supply-side effects (through the cost of imported production factors). Of estimating the model with data, Gylfason and Risager (1984) used available statistical estimates of behavioral and structural parameters from existing literature and applied them to a group of highly indebted industrial and developing countries. The results indicated that foreign debt and interest payments tend to lower the short-to-medium-run effect of devaluation on national income especially in the LDCs but

they have little impact on the generally positive effect on the current account. This study was among the first to combine the elasticity, absorption and monetary approaches into a comprehensive framework. It helped establish the link, between mechanisms and empirical assessment that has become typical in many later J-Curve studies.

A growing consensus in the literature suggests that aggregate trade analysis may inadvertently mask critical bilateral dynamics, leading to skewed or inconclusive results. This realization has prompted a shift toward examining J-Curve effects at the bilateral level, utilizing more granular data and sophisticated econometric tools. By isolating specific trade relationships, researchers have uncovered nuanced patterns that remain invisible in national aggregates, thereby highlighting the inherent risks of aggregation bias.

Early challenges to the universality of the J-Curve were noted by Rose and Yellen (1989), who investigated whether a J-Curve could be detected in twenty-five years of U.S. data. Using a nonstructural partial reduced form approach that models the real trade balance as a function of the real exchange rate and real home and foreign output, they found no statistically reliable evidence of a stable J-Curve. Their results suggested that the relationship between the real exchange rate and the trade balance was far weaker and less stable than the simple J-Curve narrative implied, and that the Marshall-Lerner condition did not hold reliably in the data. This finding was influential precisely because it showed that the J-Curve is not a universal phenomenon, and that empirical results depend heavily on model specification and the choice of exchange rate measures.

Moving beyond bilateral partners, research has also explored sectoral asymmetries. Bahmani-Oskooee and Hajilee (2009) extended this logic to the industry level, analyzing Sweden-United States trade. Their groundbreaking study revealed that J-Curve effects vary fundamentally across different product categories, with some industries showing clear J-Curves while others exhibited reverse patterns. This evidence is particularly relevant for the current study, as it suggests that the trade balance of a country like Nepal may be driven by specific commodity groups rather than a uniform national response.

The influence of external shocks and economic transitions further complicates the J-Curve narrative. Lal and Lowinger (2002) examined East Asian economies during the Asian financial crisis, finding mixed evidence that highlighted the role of exchange rate regimes and structural constraints. Similarly, Grigoryan (2015) analyzed Armenia's

transition economy, finding that the J-Curve operated in the short run but lacked long-term consistency. These studies suggest that for small, open, or transitioning economies, the J-Curve is not a static law but a dynamic response shaped by the economic environment.

Finally, a meta-analysis by Kallianiotis (2022) provided a recent empirical synthesis by testing the J-Curve effect between the United States and six trading partners—the Euro-zone, Canada, the United Kingdom, Switzerland, Japan, and Australia. Using monthly data and estimating both a GARCH-M model and a VAR model, he found that real exchange rate changes have a significant impact on the U.S. trade balance. The results supported the existence of the J-Curve, showing that a dollar depreciation worsens the trade balance in the short run before improving it over time. The study concluded that while the theoretical foundation of the J-Curve remains robust, the empirical results vary across countries, confirming that the inconsistency in previous findings is likely a product of differing methodological approaches and country-specific contexts. This provides a strong justification for the tailored, country-specific approach adopted in this research.

2.4 The NARDL Methodology: Development and Applications

The emergence of the Non-linear Autoregressive Distributed Lag (NARDL) methodology represents a pivotal advancement in J-Curve research, fundamentally altering the analysis of exchange rate-trade balance transmissions. Developed by Shin et al. (2014), the NARDL framework extends the linear ARDL bounds testing approach pioneered by Pesaran et al. (2001) to accommodate asymmetric effects within a rigorous econometric structure. This section examines the development of the NARDL model and its application to J-Curve studies, highlighting its theoretical and empirical superiority over traditional linear frameworks.

The foundation of this approach lies in the linear ARDL model introduced by Pesaran et al. (2001). This framework revolutionized cointegration analysis by permitting the inclusion of variables with differing orders of integration—specifically a mix of $I(0)$ and $I(1)$ variables—within a single coherent model. Unlike the Johansen cointegration method, which necessitates that all variables be integrated of the same order, the ARDL approach is particularly robust for small sample sizes and provides reliable inference even when integration orders are mixed or unknown. By comparing the F-statistic against critical value bounds, researchers can determine the existence of a long-run relationship without

the need for exhaustive pre-testing of unit roots, making it an ideal tool for analyzing the often-volatile trade data of developing economies.

Building upon this foundation, Shin et al. (2014) introduced a Non-linear extension that allows for the decomposition of explanatory variables into asymmetric components. The core methodological innovation involves partitioning a variable—such as the exchange rate (X_t)—into positive and negative partial sums. Specifically, $X_t^+ = \sum_{i=1}^t \max(\Delta X_i, 0)$, representing cumulative appreciation, and $X_t^- = \sum_{i=1}^t \min(\Delta X_i, 0)$, representing cumulative depreciation. This decomposition enables the model to capture three distinct types of asymmetries: asymmetric long run effects where the coefficients differ $\beta^+ \neq \beta^-$ and asymmetric short-run adjustment speeds, and asymmetric dynamic multipliers, which map the divergent adjustment paths following positive and negative exchange rate shocks. A defining feature of the NARDL specification— emphasized by Shin et al. (2014) themselves—is that the partial-sum processes and the level variables are permitted to carry different lag orders within a single regression. This property, which is the framework’s ”crucial feature,” explicitly accommodates the empirical regularity that the optimal lag length of the partial sums often differs from the optimal lag length of the dependent variable and the level regressors. Consequently, the data-driven selection of lag-0 on the partial sums (when information criteria so indicate) is a legitimate outcome of the model, not a violation of its structure, and aligns with the parsimony principle embedded in the underlying ARDL bounds-testing approach.

The NARDL specification offers four critical advantages over its linear predecessors. First, it facilitates the direct testing of asymmetry through Wald tests on coefficient restrictions, providing a statistically rigorous basis for rejecting the symmetry hypothesis. Second, by producing separate estimates for appreciation and depreciation, it allows for more nuanced and targeted policy prescriptions. Third, it can uncover J-Curve effects that remain latent in linear models due to the ”averaging” of opposing forces. Finally, the use of dynamic multipliers provides a visual and quantitative representation of how the trade balance evolves over time, offering a far more sophisticated diagnostic of the economy’s response to currency volatility than a single coefficient ever could.

The small-sample efficiency of the NARDL framework is particularly well established in the recent applied literature. Abegaz (2024), in a NARDL analysis of

Ethiopia's trade balance over 1992–2022 ($T = 31$), explicitly justifies the use of the framework on grounds that it "provides efficient and reliable estimates in case of low and limited samples" and "allows the variables to be integrated with different orders of integration, and estimated at different lag lengths in one regression model." Iqbal et al. (2021) document a similar pattern in bilateral NARDL applications to Pakistan's trade with eight partners, and the J-Curve applications surveyed in Bahmani-Oskooee and Fariditavana (2016) routinely report lag orders on the partial sums that differ from those on the level terms. These precedents indicate that the NARDL framework is the method of choice when the available sample cannot support a heavily parameterized specification, and they provide direct support for the present study's small-sample context ($T \approx 43$).

The transition from theoretical framework to empirical application was most prominently exemplified by Bahmani-Oskooee and Fariditavana (2016), who published a seminal application of the NARDL approach to J-Curve analysis in the *Open Economies Review*. By analyzing bilateral trade data between the United States and four major partners—Canada, China, Japan, and Korea—their research provided a definitive empirical case for asymmetry. Their findings were transformative in three primary dimensions. First, they uncovered significant asymmetric effects across all four country pairs, suggesting that asymmetry is a pervasive feature of trade dynamics rather than an anomaly. Second, they demonstrated that the J-Curve phenomenon was more frequently detected when asymmetry was permitted, whereas linear models often failed to identify the effect entirely. Third, the divergence in sign and magnitude between short-run and long-run coefficients underscored the critical necessity of distinguishing between these two temporal dynamics. This study effectively established the NARDL framework as the superior methodology for contemporary J-Curve testing.

This methodological shift was further corroborated by Arize et al. (2017), who extended the NARDL application to a broader sample of eight countries. Their research reinforced the notion that linear models act as a "mask" obscuring non-linear dynamics that are only revealed when depreciation is analyzed separately from appreciation. By isolating these movements, they were able to identify both short-run and long-run asymmetric effects that had remained invisible in previous aggregate linear studies. Collectively, these studies provide a powerful empirical justification for the present research; they suggest that

for an economy with the structural complexities of Nepal, a linear approach would likely result in a mis-specified model and potentially misleading policy conclusions.

In recent years, the NARDL methodology has seen pervasive adoption across global scholarship, providing robust evidence of asymmetric effects and J-Curve patterns through increasingly granular analytical lenses. The current trend in the literature suggests a decisive shift away from aggregate national analysis toward disaggregated sectoral and bilateral investigations, recognizing that "average" trade responses often obscure critical economic heterogeneities.

The importance of this sectoral approach is most evident in the work of Bahmani-Oskooee et al. (2019), who applied the NARDL framework to China-United States commodity trade. By disaggregating data by industry, they revealed that asymmetric effects vary fundamentally across different sectors of the economy. This finding serves as a critical warning against the use of aggregate data, as it demonstrates that sectoral dynamics can be so divergent that they effectively neutralize one another in national totals. Similarly, Bahmani-Oskooee et al. (2020) conducted a comprehensive analysis of Pakistan-United States trade across 41 distinct industries. Their findings reinforced the notion that the J-Curve is not a universal phenomenon; rather, its presence is highly contingent upon the industrial composition of the trade relationship, with only a subset of industries exhibiting the classic J-shaped response.

The robustness of the NARDL approach in detecting "invisible" asymmetries has also been validated across diverse economic contexts. Bahmani-Oskooee and Harvey (2017) demonstrated this through two influential studies on the economies of Singapore and Tunisia, respectively. In both instances, NARDL uncovered asymmetric J-Curve effects that had remained undetected by traditional linear models, further validating the methodology's superior sensitivity.

Of particular relevance to the Nepalese context is the recent study by Ganai and Khan (2025) on India-United States trade. Utilizing the NARDL framework, they identified significant asymmetric J-Curve effects characterized by divergent adjustment speeds for appreciation and depreciation. Given India's role as Nepal's primary trading partner and the existing currency peg, their findings provide a vital regional benchmark for understanding how asymmetry and adjustment lags operate in South Asian trade dynamics.

Collectively, these recent applications underscore the necessity of employing non-linear, disaggregated models to capture the true complexity of international trade.

A further strand of recent NARDL applications targets small, open, and pegged or managed exchange-rate regimes whose data-generating processes closely mirror Nepal's. Abegaz (2024) applies the framework to Ethiopia's trade balance over 1992–2022 and reports asymmetric short- and long-run effects in both the appreciation and depreciation segments; importantly, the study uses Zivot–Andrews unit-root tests to accommodate structural breaks and bounds testing for cointegration, and documents that the lag structure of the partial sums differs from that of the level terms—directly illustrating the lag-elasticity property emphasized in the original Shin et al. formulation. This application, together with the sectoral NARDL evidence in Albiman (2022) for Tanzania and Uganda and the regional NARDL evidence in Pertiwi and Syamad (2024) for East Java, demonstrates that the methodology is well-suited to developing-country contexts in which the trade-balance response to exchange-rate shocks is shaped by structural rigidities, import dependence, and a constrained set of adjustment instruments—all features that characterize the Nepalese economy.

2.5 Country-Specific NARDL Studies from Developing Regions

The recent application of the NARDL framework across diverse economic contexts provides a robust empirical basis for investigating the Nepalese trade balance, particularly through the lens of regional peers and developing economies. The evidence from South Asia is especially salient. Bhat and Bhat (2021) utilized an asymmetric Non-linear cointegration approach to analyze India's trade balance, examining whether currency appreciation and depreciation have different effects. Their study found no evidence of the J-Curve phenomenon in the case of India, but it did uncover significant asymmetric effects. In the short run, currency appreciation deteriorates the trade balance while currency depreciation improves it. In the long run, a similar pattern emerges, though only the impact of currency depreciation is statistically significant. Given the systemic importance of the India-Nepal trade relationship and the existing currency peg, the asymmetric transmissions observed in the Indian economy may have direct spillover effects on Nepal's trade dynamics. Similarly, Dey et al. (2025) examined Bangladesh's bilateral trade with the United States, finding evidence of asymmetric effects that challenged the traditional J-

Curve hypothesis. The fact that a regional peer like Bangladesh exhibits these non-linearities underscores the necessity of moving beyond symmetric models when analyzing South Asian trade.

Beyond the immediate region, the NARDL methodology has proven critical in revealing dynamics that traditional linear models consistently overlook. A pivotal study by Iqbal et al. (2021) on Pakistan's bilateral trade with eight major partners demonstrated that while linear ARDL models showed no evidence of a J-Curve, the NARDL approach revealed clear J-Curve patterns in trade with Malaysia, China, and the United States. This finding provides a powerful methodological justification for the present study: it suggests that the "absence" of a J-Curve in previous Nepalese research may be a result of model misspecification rather than a lack of economic phenomenon. This is further supported by Nasir and Leung (2021), whose NARDL analysis of US quarterly data from 1994Q1 to 2018Q1 confirmed significant short and long-run asymmetries between the real effective exchange rate and the US trade balance, providing evidence of an asymmetric J-Curve.

The applicability of NARDL to other developing and transition economies further validates its use in the Nepalese context. In Africa, Abegaz (2024) identified asymmetric short- and long-run effects in Ethiopia's trade balance, while Albiman (2022) found significant asymmetric effects and short-run J-Curve evidence in Tanzania and Uganda. Conversely, Pertiwi and Syamad (2024) found that while asymmetry existed in East Java's trade with China, Japan, and the US, a full J-Curve did not materialize. Collectively, these diverse findings—ranging from the presence of a full J-Curve to the existence of asymmetry without a J-Curve—highlight the empirical uncertainty that necessitates a rigorous, non-linear investigation into the specific structural rigidities of Nepal.

2.6 South Asian and Nepal-Specific Evidence

The South Asian region provides a critical empirical benchmark for this study, as the trade dynamics of neighboring economies—particularly India and Pakistan—offer significant insights into the potential for J-Curve effects in Nepal. The relevance of these regional studies is rooted in shared structural characteristics: a heavy reliance on primary commodity exports, similar price elasticities, and comparable trade dependencies on a few major global partners. Consequently, the findings from this region serve as a proxy for the economic frictions that likely govern Nepal's own trade balance.

Early efforts to establish the J-Curve within this regional context were led by Arora et al. (2003), whose bilateral analysis of India's trade with seven major partners used quarterly data from 1977 to 1998 and an ARDL model to test for the J-Curve. Their results provided only limited evidence: they found no long-run relationship between India's trade balance and the real exchange rate, indicating that rupee depreciation had no long-run effect. A long-run improvement in the trade balance was found only for Australia, Germany, Italy, and Japan. This work nonetheless established an early empirical benchmark for the region and laid the groundwork for more sophisticated later studies.

The evolution toward non-linear analysis is best exemplified by the work of Bhat and Bhat (2021). Their application of the NARDL framework to India's trade balance revealed the existence of asymmetric Non-linear cointegration, proving that the impact of exchange rate movements in India is fundamentally asymmetric. For Nepal, this finding is particularly salient; given the tight monetary peg and the deep systemic integration between the Nepalese and Indian economies, the asymmetric transmissions observed in India likely mirror the dynamics at play in Nepal.

Further methodological justification is provided by Iqbal et al. (2021) in their analysis of Pakistan's bilateral trade. Their study yielded a critical insight: in several cases—specifically trade with Malaysia, China, and the United States—the J-Curve remained invisible under linear ARDL models but became statistically significant under the NARDL framework. This result provides a powerful rationale for the present study's approach. It suggests that the lack of J-Curve evidence in previous Nepalese literature may not be a reflection of the economy's actual behavior, but rather a consequence of employing linear models that are blind to asymmetric responses.

While the global and regional literature on the J-Curve is extensive, Nepal-specific research remains relatively scarce, providing a fragmented but essential context for the present study. This section provides a critical synthesis of existing Nepalese studies, evaluating their methodologies and identifying the empirical voids that this research seeks to fill.

The most foundational contribution to this discourse is the work of Chaulagai (2015), the most comprehensive attempt to date to map the J-Curve in Nepal, utilizing a wide array of tools including Johansen cointegration, Vector Autoregression (VAR), and

linear ARDL bounds testing. His primary finding was a complete absence of the J-Curve phenomenon; the study revealed no cointegration between the trade balance and either the Nominal Effective Exchange Rate (NEER) or the Real Effective Exchange Rate (REER). Instead, Chaulagai proposed the existence of an "L-Curve"—a stagnant response suggesting that Nepal's trade balance is largely insensitive to exchange rate fluctuations, likely due to the inelastic demand for essential imports.

However, from a methodological standpoint, this study relied exclusively on linear frameworks. By failing to test for asymmetric effects, the "L-Curve" finding may not be a reflection of economic reality, but rather a result of linear models averaging out divergent asymmetric responses. This conceptual oversight constitutes the primary gap that the present study addresses.

Complementing this, Pathak (2020) focused on the underlying drivers of these dynamics by estimating trade elasticities and testing the Marshall-Lerner condition. This work provided the necessary theoretical grounding to understand why the J-Curve might fail to materialize in Nepal, suggesting that the lack of export responsiveness and the necessity of critical imports hinder the transmission of exchange rate changes to the trade balance. More recently, emerging evidence suggests that the "linear" assumption may indeed be flawed. Bishwakarma et al. (2024) employed the ARDL cointegration approach to analyze import demand sensitivity and found evidence of asymmetric responses to exchange rate fluctuations. This provides the first strong empirical hint that asymmetry is a relevant feature of Nepal's trade. Similarly, Adhikari (2020) examined Nepal-India bilateral trade and uncovered asymmetric effects that warrant a more rigorous investigation. Given that India is Nepal's dominant trading partner, his findings suggest that the "L-Curve" identified in aggregate studies may disappear when a non-linear lens is applied to specific bilateral relationships.

Methodologically, the NARDL framework has also begun to be applied directly to Nepal and to the Nepal-India bilateral relationship. A working-paper application by an author affiliated with the Nepal Rastra Bank deploys the Shin et al. (2014) NARDL specification to model the equilibrium real exchange rate of Nepal and India jointly, and reports that, while short-run anomalies exist, neither currency exhibits a persistent medium- to long-run misalignment—a finding that is consistent with the peg-induced short-run

symmetry narrative that the present study tests explicitly. This direct precedent, together with the Ethiopia application of Abegaz (2024) and the Pakistan application of Iqbal et al. (2021), demonstrates that the NARDL methodology is the appropriate tool for the small-sample, pegged-regime, developing-country context that defines the present study.

Together, these studies establish a clear trajectory: while early research suggested an insensitive "L-Curve" through linear modeling, recent evidence points toward a more complex, asymmetric reality. The present study therefore seeks to resolve this contradiction by applying the NARDL framework. This study does not presume that the J-Curve exists in Nepal; rather, it tests whether the absence of a J-Curve in previous linear studies is a genuine structural finding or a consequence of model misspecification. If the NARDL framework confirms an L-shaped response even after permitting asymmetric effects, this would strengthen Chaulagai's (2015) conclusion that Nepal's trade balance is structurally insensitive to exchange rate movements. Conversely, if NARDL reveals asymmetric J-Curve patterns invisible to linear models, this would suggest that the L-curve finding was indeed a methodological artifact.

2.7 Research Gap and Its Importance for Nepal

This comprehensive literature review has examined the theoretical foundations, empirical evidence, and methodological framework for understanding the J-Curve phenomenon with extraordinary detail. The review has revealed several key insights that guide the present research. First, the J-Curve has strong theoretical support based on the Marshall-Lerner condition and adjustment dynamics. Second, the empirical evidence is mixed, with significant variation across countries, time periods, and methodological approaches. Third, the NARDL methodology has emerged as the preferred approach for testing asymmetric effects that linear models may miss. Fourth, in South Asia and Nepal specifically, the evidence is limited. Chaulagai's (2015) study found no linear J-Curve in Nepal, but no study has applied the asymmetric NARDL methodology.

The literature review has identified several critical gaps that the present research addresses. First, no known study has applied NARDL to Nepal's aggregate trade balance to test for asymmetric J-Curve effects. Chaulagai (2015) used linear ARDL and found no effect, but linear methods may miss asymmetric effects that NARDL can detect. Second, the time period in existing Nepal studies (up to 2013-2015) predates significant exchange

rate dynamics. Nepal has experienced substantial exchange rate volatility since 2015, and the data period 1980-2024 captures different economic conditions. Third, no Nepal study has explicitly tested the asymmetry hypothesis—whether exchange rate depreciation and appreciation have different effects on Nepal’s trade balance. Fourth, most Nepal studies used Nominal Effective Exchange Rate (NEER), while international NARDL studies typically use Real Effective Exchange Rate (REER), which accounts for inflation differentials.

The studies by Iqbal et al. (2021) on Pakistan and Bishwakarma et al. (2024) on Nepal provide direct evidence that NARDL can reveal effects that linear models miss. In Pakistan’s case, linear ARDL showed no J-Curve but NARDL revealed J-Curve evidence for several trading partners. In Nepal’s case, Bishwakarma et al. (2024) found asymmetric responses to exchange rate changes using linear ARDL, suggesting that asymmetry is present but was not fully tested.

First, Nepal has experienced persistent trade deficits for decades, and understanding whether exchange rate adjustments can improve the trade balance is essential for policy. If the J-Curve exists—even asymmetrically—currency depreciation could be an effective policy tool. If it does not exist, policymakers should explore alternative approaches. Second, Nepal is highly dependent on imports of essential goods including petroleum products, pharmaceuticals, and machinery. Understanding how these imports respond to exchange rate changes is crucial for managing the balance of payments. Third, Nepal’s close trading relationship with India—accounting for approximately 60% of total trade—makes understanding bilateral dynamics essential. Fourth, with the extended data period (1980-2024), this study captures post liberalization dynamics (1990s), the global financial crisis (2008-2009), and recent exchange rate movements, providing a comprehensive analysis.

This chapter has presented an incredibly comprehensive literature review that establishes the theoretical foundation, surveys the empirical evidence, and defines the methodological framework for the present study. The review has demonstrated that the J-Curve phenomenon has been extensively studied globally using various econometric approaches, but that Nepal-specific evidence remains limited. The key insight from the literature is that asymmetry—often missed by linear models—may reveal effects where

linear models find none. For Nepal, existing linear studies found no J-Curve, but no study has applied the asymmetric NARDL methodology. The present research addresses this gap directly by applying NARDL to Nepal's trade data from 1980-2024, testing for asymmetric J-Curve effects using Wald tests, and providing policy recommendations based on the findings.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Research Design

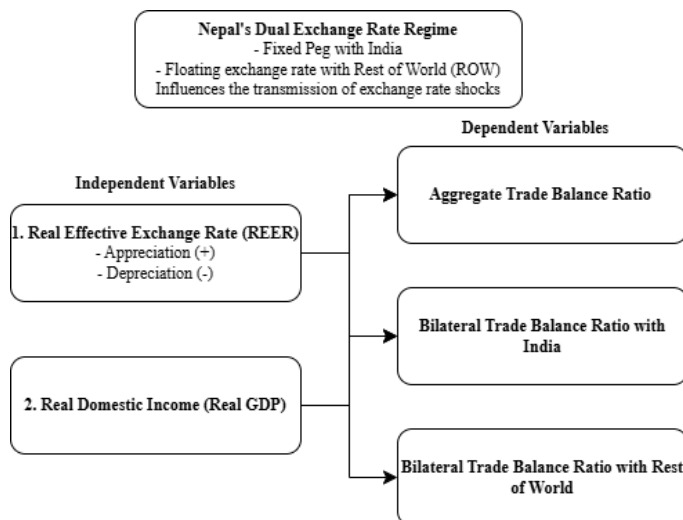
The study uses a quantitative time-series approach with annual data for Nepal from 1980 to 2024. The analysis is done in three stages each applied separately to three trade balance ratios: the Trade Balance Ratio (TBR total) trade balance ratio with India (TBR India) and trade balance ratio with the Rest of the World (TBR ROW). This helps capture both non-linear patterns in the data.

First a linear ARDL bounds-testing model is used as the baseline specification. This sets the foundation, for the analysis. Second an Error Correction Model (ECM) is derived from the ARDL model to look at short-run adjustments and how quickly the system returns to long-run equilibrium. Third a Non-linear ARDL (NARDL) model is applied. This model separates real exchange rate changes into appreciation and depreciation parts. It allows the ECM to be re-estimated within a framework. This three-stage design lets all three hypotheses be tested together in one coherent modeling structure.

3.2 Conceptual Framework

Figure 1

Conceptual Framework



This study is based on combining two related viewpoints. First it uses the elasticity method to check if real currency depreciation helps improve the trade balance. It looks at whether the long term exchange rate coefficient matches the Marshall-Lerner prediction. That prediction says the total of export and import price elasticities should be greater than one. Second it uses the absorption method. It includes domestic income as a control variable. This is because when domestic income grows it can increase the trade deficit. That happens because higher income leads to demand for imports regardless of what happens to the exchange rate. The study also takes into account the structural context of Nepal's dual exchange rate system. In this system movements in the effective exchange rate are mainly caused by differences in inflation compared to floating global currencies (the Rest of the World or ROW). However the exchange rate, with India is fixed because of the peg.

3.3 Tools and Techniques of Data Collection

To understand these dynamics the research uses a three-stage framework that is applied separately to three trade balance ratios: the aggregate trade balance (TBR total) the bilateral trade balance with India (TBR India) and the bilateral trade balance with the Rest of the World (TBR ROW). This disaggregation is vital because Nepal's two regimes can create exchange-rate transmission mechanisms; the fixed peg with India could dampen adjustment while the floating REER-driven trade, with the Rest of the World could show a classic J-curve and asymmetric behavior. If these two trade balance ratios were combined into one trade balance, regime-specific effects might cancel each other out leading to misleading conclusions.

For each of the three variables the same three-stage sequential procedure is followed. First the Linear ARDL bounds-testing approach of Pesaran et al. (2001) is estimated as the baseline. This is done to identify the run co-integrating relationship and to provide a symmetric benchmark. Second an Error Correction Model (ECM) is derived from the ARDL. This step captures the short-run dynamics and the speed of adjustment to run equilibrium. This ECM is the vehicle for testing the J-Curve, which is Hypothesis 1. Third the Non-linear ARDL (NARDL) framework of Shin et al. (2014) is estimated where this framework decomposes the exchange rate into positive partial sums (appreciation) and negative partial sums (depreciation). From this a NARDL-ECM is derived to test for

effects, which relates to Hypothesis 2. The independent variables used across all three regimes are the REER and Nepal's real domestic income (GDP). Foreign income which was considered under the imperfect-substitutes framework however, it is excluded from the specification on grounds of parsimony. Including it in pilot estimations led to coefficient instability in the small-sample context ($T \approx 43$). This finding is consistent with the warning in Abegaz (2024) and Pertiwi and Syamad (2024), on the degrees-of-freedom cost of adding regressors in small-sample NARDL applications.

This three-stage multi-level design helps the study to capture both non-linear dynamics of Nepal's trade balance adjustment to exchange rate shocks. It also reveals whether the J-Curve and asymmetry appear or disappear depending on the regime whether that regime is India or the rest of the world. This approach is a shift from earlier Nepal research, which only used linear aggregate models.

For this research we extracted data, from NRB publications and the World Bank WDI database.

Data Retrieval: Historical data points were extracted from archives of the NRB quarterly economic bulletin (Exports M NPR Imports M NPR, Exp India M NPR, Exp ROW M NPR, Imp India M NPR, Imp ROW M NPR) and from the World Bank WDI database (GDP USD, Exchange rate npr).

Variable Construction and Data Transformation: To make the NARDL model reliable we transformed the data. These transformations helped to stabilize variance.

1. Trade Balance Ratio (TBR): Rather than using a simple difference, which is exports minus imports this study uses the Trade Balance Ratio, which is calculated as the ratio of exports, to imports. This method reduces the effect of inflation. Gives a normalized way to look at trade performance. Three different Trade Balance Ratio series are created as follows

$$TBR_{total} = \frac{Exports_M_NPR}{Imports_M_NPR}$$

$$TBR_{India} = \frac{Exp_India_M_NPR}{Imp_India_M_NPR}$$

$$TBR_{ROW} = \frac{Exp_{ROW_M_NPR}}{Imp_{ROW_M_NPR}}$$

Each series is changed using logarithms so that the coefficients can be understood as elasticities.

2. Log-Transformation and Elasticity: To handle non-linearity and make the variance more stable both the TBR series and the real GDP series are turned into logarithms. This step helps to make the coefficients easier to understand. After the transformation the coefficients show elasticities. That means the elasticities reflect the percentage change in the trade balance ratio when there is a 1 percent change in the exchange rate or, in income.

3. The Real Effective Exchange Rate (REER): The REER is sourced from the Bruegel database, which serve as the independent variable. This measure shows how the Nepalese rupee performs against a basket of currencies with weights based on trade patterns. It is adjusted for differences in inflation, between Nepal and its trading partners giving a picture of real international competitiveness. The REER is also log-transformed to make it easier to work with in analysis.

4. Partial Sum Decomposition (NARDL Process): Following the NARDL methodology (Shin et al., 2014) the log-transformed REER series is broken down into two asymmetric parts to show how currency changes affect things differently.

- Positive Partial Sum (REER+): This is the total of all changes in the log REER. It shows times when the real currency got stronger.
- Negative Partial Sum (REER-): This is the total of all changes in the log REER. It shows times when the real currency got weaker.

This breakdown is the part of the asymmetric analysis. It helps the model see if a 1% increase in the currency has an effect on the trade balance, than a 1% decrease. The same broken down parts are used for all three TBR models (India and ROW) to make sure they can be compared.

3.4 Tools & Techniques of Data Analysis and Model Specification

The empirical analysis is carried out in three steps that come one after the other. The first step uses a Linear ARDL bounds-testing model as introduced by Pesaran et al. (2001) to set the baseline. The second step applies an Error Correction Model or ECM to

examine short-run changes and how quickly the system returns to equilibrium. The third step uses the NARDL framework from Shin et al. (2014) which allows for effects. Each of these three steps is applied separately to three trade balance ratios. The first is the trade balance ratio, referred to as TBR_total. The second is the trade balance ratio with India called TBR_India. The third is the trade balance ratio with the Rest of the World known as TBR_ROW. The same two independent variables— effective exchange rate (REER) and real GDP—are used in every model to keep the results comparable, across all specifications. All estimations are done using EViews 12. The partial-sum decomposition and the NARDL setup follow the steps described in Shin et al. (2014). Diagnostic tests and multiplier plots are also generated within the same EViews environment. To make sure the findings are sound and trustworthy the study follows a six-step process that is applied in sequence.

Step 1: Stationarity and Integration Analysis

Before starting the model estimation we check how each variable behaves over time. This includes the TBR series, the effective exchange rate (REER) and real GDP. We use two tests: the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test. These tests help us see if the variables are stationary or not. The goal is to make sure none of them are integrated of order two which means I(2). Because both the ARDL and NARDL models only work properly when the variables are either stationary at levels meaning I(0) or become stationary after differencing, which means I(1). If a variable is I(2) these models can give results. So checking this step carefully is important, for getting estimates.

Step 2: Linear ARDL Model Estimation (Baseline)

The basic linear ARDL model (p, q, r) from Pesaran et al. (2001) is calculated individually for each TBR series (total, India rest of the world) using REER and real GDP as factors. The run values that are calculated and the F-statistic for checking limits create the linear base that is used to compare the results from the asymmetric NARDL.

The general $ARDL(p, q, r)$ specification for any given TBR series is:

$$\Delta TBR_t = \alpha_0 + \sum_{i=1}^p \alpha_i \Delta TBR_{t-i} + \sum_{j=0}^q \beta_j \Delta REER_{t-j} + \sum_{v=0}^r \gamma_v \Delta GDP_{t-v} + \delta_1 TBR_{t-1} + \delta_2 REER_{t-1} + \delta_3 GDP_{t-1} + \epsilon_t \quad (1)$$

Variable Definitions.

- Δ denotes the first-difference operator.

- TBR = Trade Balance Ratio (aggregate, India, or ROW, in natural logs).
- $REER$ = Real Effective Exchange Rate (in logs; symmetric).
- GDP = Real domestic income (Nepal, in logs).
- p, q, r = optimal lag lengths selected by SIC, with the partial-sum lag order (q) permitted to differ from the dependent-variable lag order (p) in line with the Shin et al. (2014) NARDL specification.
- ε_t = white-noise error term.

Remark on the exclusion of foreign income. The imperfect-substitutes model provides a reason to include a foreign-income control. In tests the foreign-income regressor—India’s real GDP for the India model and world real GDP for the aggregate and ROW models—was added. In all three regimes the coefficient for this variable did not show statistical significance in pilot test. Also it led to long-run REER coefficients when using the parsimonious lag structure selected by information criteria. Because of these issues the foreign-income regressor is left out of the model. This decision is based on the principle of parsimony. The theoretical reason, for including the exchange rate as the price-effect channel is still valid. At the time the domestic-income control continues to handle the absorption-effect channel.

Step 3: Bounds Testing for Long-Run Cointegration

We check whether a stable long-run relationship exists between each TBR series, REER and real GDP. First we use the linear ARDL model. Then we use the NARDL model following the Pesaran et al. (2001) Bounds testing method. We look at the F-statistic that comes from testing whether the lagged level variables ($\delta_1 = \delta_2 = \delta_3 = 0$) are jointly significant. We compare this F-statistic to the bound $I(0)$ and the upper bound $I(1)$. If the F-statistic is higher than the bound we reject the null hypothesis that there is no cointegration and we say that the variables have a long-run equilibrium. If the F-statistic lies, between the two bounds the result is inconclusive.

Step 4: Error Correction Model (ECM) Estimation

Once cointegration is confirmed the run coefficients are found by normalizing the ARDL estimates. Then the model is rewritten in error-correction form. The error-correction model or ECM shows the short-run movements. The lagged error-correction term, (ECT_{t-1}) reveals how fast the system returns to run equilibrium. The sign, statistical

significance and size of this term directly tell us how quickly adjustment happens. The ECM is the tool used to test Hypothesis 1 which is about the J-Curve. The pattern of signs, for the run and long-run depreciation coefficients can be read straight from the ECM.

$$\Delta TBR_t = \alpha_0 + \sum_{i=1}^p \alpha_i \Delta TBR_{t-i} + \sum_{j=0}^q \beta_j \Delta REER_{t-j} + \sum_{v=0}^r \gamma_v \Delta GDP_{t-v} + \lambda ECT_{t-1} + \epsilon_t \quad (2)$$

where ECT_{t-1} is the lagged residual from the long-run co-integrating relationship.

The discussion on lag order has effects on the Wald-based test of the short-run restriction on the partial-sum coefficients. When the information criteria choose lag-0 for both $\Delta REER^+$ and $\Delta REER^-$, the run Wald restriction $H_0^{SR}: \beta_j^+ = -\beta_j^-$ becomes a simple comparison between two contemporaneous coefficients. In this case it no longer serves as a test of the asymmetric short-run adjustment mechanism. However the run restriction $H_0^{LR}: \theta^+ = \theta^-$, remains fully identified. This is because the run coefficients are derived by normalizing the level-equation parameters and they stay well-defined even when the short-run partial sums are at lag-0. As a result the study focuses on the run Wald test as the main test for Hypothesis 2. It also clearly limits the asymmetry test to the long-run relationship. The short-run asymmetry restriction is recognized as not being separately identifiable, under the specification that the data support. This is not treated as a flaw but rather as a substantive finding. It aligns with the narrative of peg-induced short-run symmetry.

Step 5: NARDL Model and Asymmetry Testing

The REER series is decomposed into positive (appreciation) and negative (depreciation) partial sums:

$$REER_t^+ = \sum_{j=1}^t \max(\Delta REER_j, 0), REER_t^- = \sum_{j=1}^t \min(\Delta REER_j, 0) \quad (3)$$

The NARDL model is then calculated for each TBR series. The NARDL-ECM form (Equation 4 is the main way to look at the data. It helps to see how the effects change in the term and also shows how things correct themselves over time. All of this is done in one regression.

$$\Delta TBR_t = \alpha_0 + \sum_{i=1}^p \alpha_i \Delta TBR_{t-i} + \sum_{j=0}^q \beta_j^+ \Delta REER_{t-j}^+ + \sum_{j=0}^r \beta_j^- \Delta REER_{t-j}^- + \sum_{v=0}^r \gamma_v \Delta GDP_{t-v} + \lambda ECT_{t-1} + \epsilon_t \quad (4)$$

The corresponding long-run asymmetric form is:

$$TBR_t = \theta_0 + \theta_+ REER_t^+ + \theta_- REER_t^- + \theta_{GDP} GDP_t + \mu_t \quad (5)$$

where θ^+ and θ^- capture the long-run asymmetric effects of appreciation and depreciation, respectively.

Hypothesis Testing

The three hypotheses are tested as follows.

Hypothesis 1 (J-Curve effect): A J-Curve is confirmed for a given regime (India or ROW) if in either the symmetric ECM (2) or the NARDL-ECM (4) the short-run coefficient on real depreciation (β_j^-) is significantly negative while the long-run coefficient (θ^-) is significantly positive. Both ECMs are reported to allow comparison. I notice that the J-Curve interpretation is, however conditional on cointegration being established by the bounds test (Step 3); for regimes in which the bounds F-statistic does not exceed the critical bound the long-run J-Curve sign pattern is not interpretable and the test outcome is recorded as "no cointegration, J-Curve not applicable." This conditionality is reported alongside the coefficient estimates, in the results chapter.

Hypothesis 2 (asymmetry): The Wald test is applied to the NARDL long-run coefficients to test the null of long-run symmetry, following Shin et al. (2014):

$$H_0^{LR}: \theta^+ = \theta^- \text{ (long run)} \quad (6)$$

The long-run Wald statistic is reported alongside its F and χ^2 equivalents. Rejection of the restriction at the 5% level is taken as evidence of long-run asymmetry. As discussed in Step 4, the short run Wald restriction on the partial-sum coefficients is not separately identified under the parsimonious specification supported by the data and is therefore not reported as a standalone test. Dynamic multipliers—the asymmetric time-path responses of TBR to positive and negative REER shocks—are plotted using the NARDL dynamic multipliers, providing a visual diagnostic of how the trade balance evolves over time under appreciation versus depreciation, and serve as a graphical complement to the formal Wald test.

Hypothesis 3 (regime-dependent asymmetry): This hypothesis is evaluated by looking at the long-term Wald results and whether the India model and the rest of the world model are connected.

Step 6: Diagnostic Checking

For each estimated model including linear ARDL, ECM, NARDL and NARDL-ECM. Applied to the data India and the rest of the world (ROW) a set of diagnostic tests is carried out. The Breusch–Godfrey Lagrange Multiplier test is used to check for serial correlation in the residuals. This test goes up to lag 2. Next, the Jarque–Bera test checks whether the residuals follow a distribution. Then the Breusch–Pagan–Godfrey test is applied to detect heteroscedasticity. This is especially important because the sample size is small and the NARDL inference can be sensitive to changes in error variance. The CUSUM and CUSUMSQ tests are used to assess parameter stability over the sample period. These tests use the 5% bounds as reference points. When the CUSUM or CUSUMSQ lines cross these bounds it suggests a shift in the model parameters. These shifts are discussed in the context of known breaks. The Ramsey RESET test is run to check for functional-form misspecification. This helps confirm that the model is correctly specified. Lastly dynamic multiplier plots from the NARDL model are generated. These plots show how the trade balance responds over time to a unit and a unit negative shock in the real effective exchange rate (REER). The cumulative responses are traced across the forecast horizon. These plots provide a way to detect asymmetry and they complement the formal Wald test, for asymmetry.

CHAPTER IV

DATA ANALYSIS AND DISCUSSION

This chapter presents the analysis and discussion of the findings that come from the sequential Linear ARDL, Error Correction Model (ECM) and Non-linear ARDL (NARDL) frameworks. The chapter start with a descriptive analysis at the data to give a clear picture of Nepal's trade dynamics how the exchange rate changes and how the economy grows from 1980 to 2024. Next the chapter proceeds with pre-estimation unit root tests and a three-stage analysis process and run each part separately for the trade balance, the bilateral trade balance with India and the trade balance, with the Rest of the World (ROW). At the end the chapter combine the findings with the theoretical literature.

4.1 Data Trend and Descriptive Analysis

Before starting econometric estimations it is essential to understand the historical path of the key variables. This part talks about the trends of the Trade Balance Ratios (TBR) the Real Effective Exchange Rate (REER) and Nepal's Real GDP from 1980 to 2024.

Table 1

Descriptive Statistics

Variable	Std.							JB-p-	
	Mean	Dev.	Min	Median	Max	Skewness	Kurtosis	JB	value
ln total tbr	-0.66	0.26	-1.18	-0.57	-0.32	-0.59	-0.98	4.42	0.11
ln india tbr	-0.68	0.27	-1.24	-0.68	-0.27	-0.24	-1.09	2.68	0.26
ln row tbr	-0.64	0.23	-1.12	-0.59	-0.23	-0.14	-0.96	1.89	0.39
ln reer	2.03	0.07	1.92	2.04	2.16	0.10	-1.42	3.86	0.14
ln nepal gdp	9.92	0.43	9.29	9.78	10.63	0.32	-1.36	4.21	0.12
ln india gdp	11.86	0.43	11.27	11.71	12.59	0.25	-1.46	4.45	0.11
ln row gdp	14.48	0.33	13.94	14.41	14.96	-0.17	-1.24	3.09	0.21

Note. $N = 45$ for all variables. All variables are in natural logarithms.

JB = Jarque-Bera test statistic for normality.

Source: calculations based on appendix 2

Table 1 shows the statistics for the log-transformed variables used in this study covering the 1980–2024 period ($T = 45$). The mean and median values for the three trade

balance ratios are all negative. This statistically confirms the core premise of this thesis: Nepal has sustained a chronic trade deficit across aggregate, bilateral (India) and floating (ROW) regimes over the past four decades. The standard deviations of the trade balance ratios indicate volatility reflecting the economy's susceptibility, to external shocks and shifting domestic absorption capacities over time.

The way the variables are spread out supports the method we used. The skewness numbers for the variables are not equal to zero, which shows that the distributions are not balanced. Also the kurtosis numbers are not equal to 3, which's the value for a normal distribution. This means there are tails—so extreme events like the 2015 earthquake or the COVID-19 pandemic are included in the data. This is officially proven by the Jarque-Bera test results and their p-values, which show that the idea of a distribution is not true for most of the data, at regular levels of significance.

The unusual changing and uneven patterns in these economic time series give a lot of real-world support for two choices in how to work with the data. First it supports using logarithms to make the data more stable and to look at the results in terms of how much one thing changes when another changes. Second and more important the way the data is different on each side supports using the Non-linear ARDL (NARDL) method. Unlike models that act like changes are even and follow a normal pattern the NARDL method works well when the data isn't normal and is made to see how big changes in one direction are different from changes, in the other direction.

4.1.1 Trend of Trade Balance Ratio (TBR)

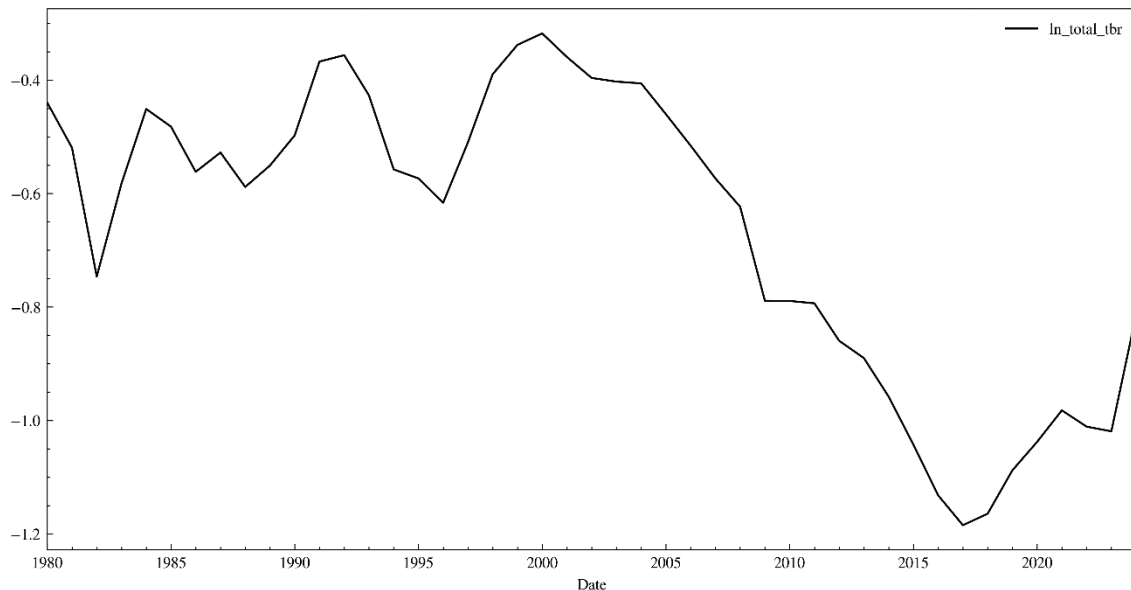
The core dependent variables in this study are the trade balance ratios, the India trade balance ratios and the Rest of the World trade balance ratios measured as the ratio of exports to imports. A ratio below one indicates a trade deficit.

Figure 2 shows the trajectories of Nepal's Trade Balance Ratios (TBR) in natural logarithmic form from 1980 to 2024. The TBR is calculated as the ratio of exports to imports; therefore a value of zero indicates a trade (exports equals imports) a positive value indicates a trade surplus whereas a negative value indicates a trade deficit. A downward slope signifies a widening deficit while an upward slope signifies improvement. By disaggregating the aggregate trade balance ratios into trade with India and trade, with the Rest of the World distinct structural divergences become visible.

1. Aggregate Trade Balance (Total TBR):

Figure 2

Aggregate trade balance in log form



Source: Nepal Rastra Bank

Over the forty years the aggregate TBR has shown a steady and worsening deficit. In the 1980s it was close to zero meaning trade was balanced. Starting in the mid-1990s the number fell sharply, kept falling. The fall got faster during the 2000s reached about – 1.2 in 2015. A log value of –1.2 means that Nepal’s total imports were 3.3 times bigger than its total exports. After 2015 there was a rise but the overall ratio is still very negative. This pattern shows that after the 1990 liberalization the Nepalese economy changed. Remittance-driven import demand grew faster, than the country’s ability to produce exports.

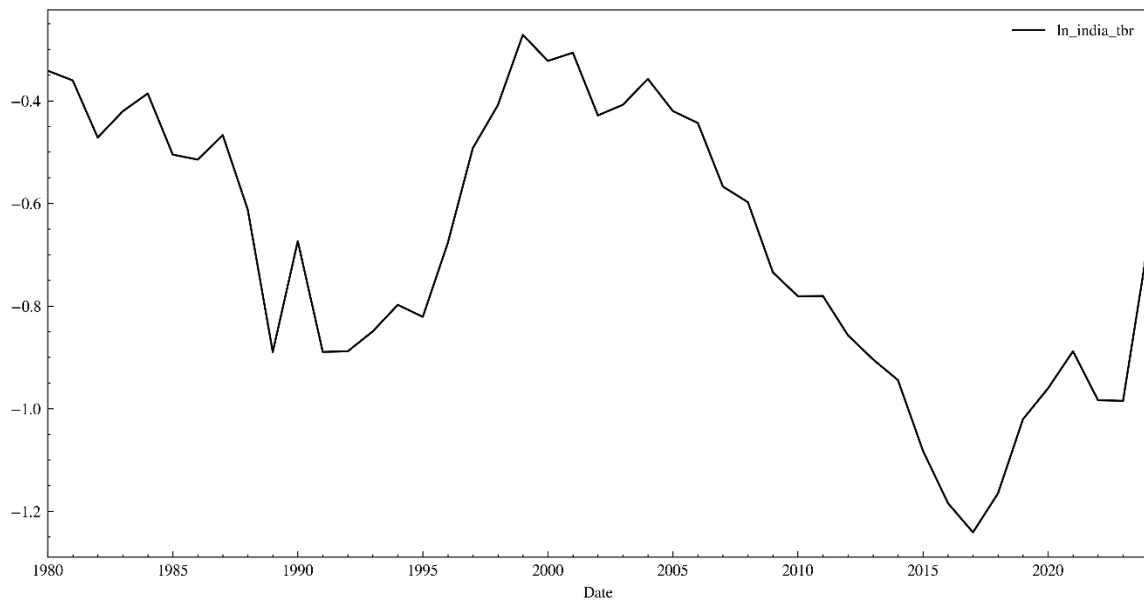
2. India Trade Balance (India TBR):

The trade balance between Nepal and India shows a deficit over the entire period staying mostly between -0.4 and -0.8. Unlike the trade balance the trade balance with India displays clear cycles rather than a steady decline. Since the Nepalese Rupee is fixed to the Rupee exchange rate changes do not affect the numbers. This means the ups and downs in the trade balance are not caused by currency shifts. Instead they come from economic factors. These include changes in crop production, agricultural cycles and shifts in output. The deficit stays high because Nepal depends heavily on India for imports, like fuel and

basic manufactured goods. At the time Nepalese exports struggle to compete in the Indian market. This makes the trade imbalance a standing issue.

Figure 3

India trade balance in log form



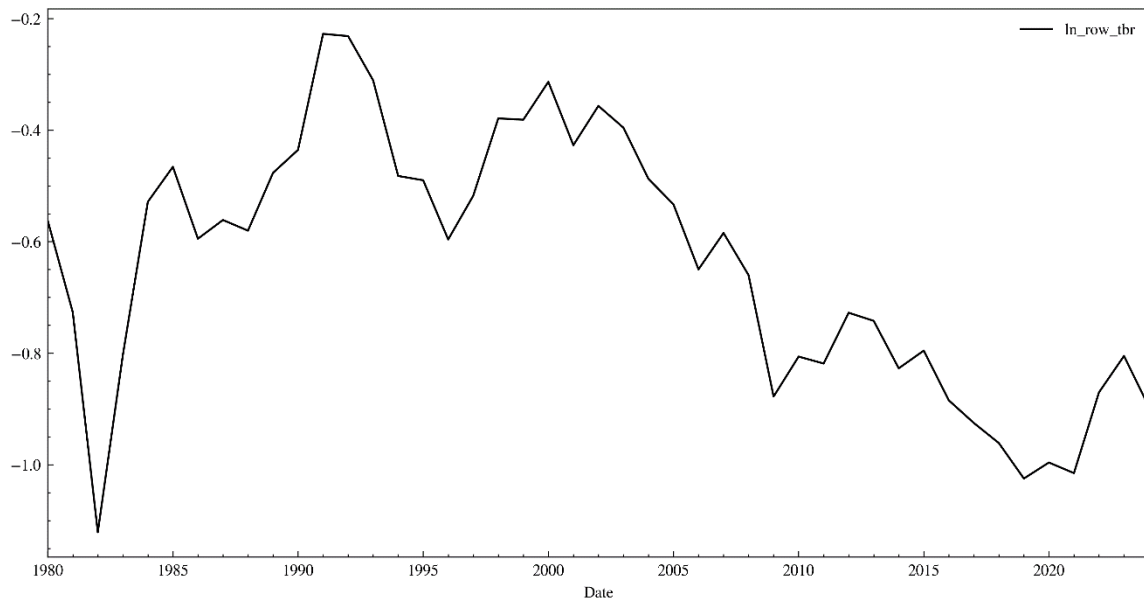
Source: Nepal Rastra Bank

3. Rest of the World Trade Balance (ROW TBR):

The trade balance with the Rest of the World stayed in a deficit during the whole sample period. The trade balance shows phases of volatility and worsening conditions. In the 1980s the trade balance with the Rest of the World saw swings. The trade balance oscillated between a deficit of about -0.4 and deeper troughs near -1.1 . From the 1990s onward the trade balance with the Rest of the World moved in a downward direction. By the post-2010 period the trade balance with the Rest of the World had become deeply entrenched with the ratio below -1.0 . This severe and continuous decline in the trade balance with the Rest of the World matches the depreciation of the NPR against hard currencies the drop, in traditional export industries as domestic production costs rose and the removal of international textile quotas.

Figure 4

Rest of the World trade balance in log form



Source: Nepal Rastra Bank

4. Real Effective Exchange Rate (REER):

The REER shows how strong the Nepalese Rupee (NPR) is compared to a group of currencies. This group is adjusted for inflation in Nepal and in other countries. When the REER goes up it means the Nepalese Rupee is getting stronger in terms. This makes Nepalese exports more expensive and less competitive in the world. When the REER goes down it means the Nepalese Rupee is getting weaker in terms. This makes Nepalese exports cheaper and more competitive in the world.

The REER started with a movement. It reached a point around 1983. This shows a time when the Nepalese Rupee was strong. From the middle of the 1980s to the middle to late 1990s the REER dropped a lot. It reached its point in the middle of the 1990s. This long period of weakness should have helped Nepalese exports become more competitive during that time.

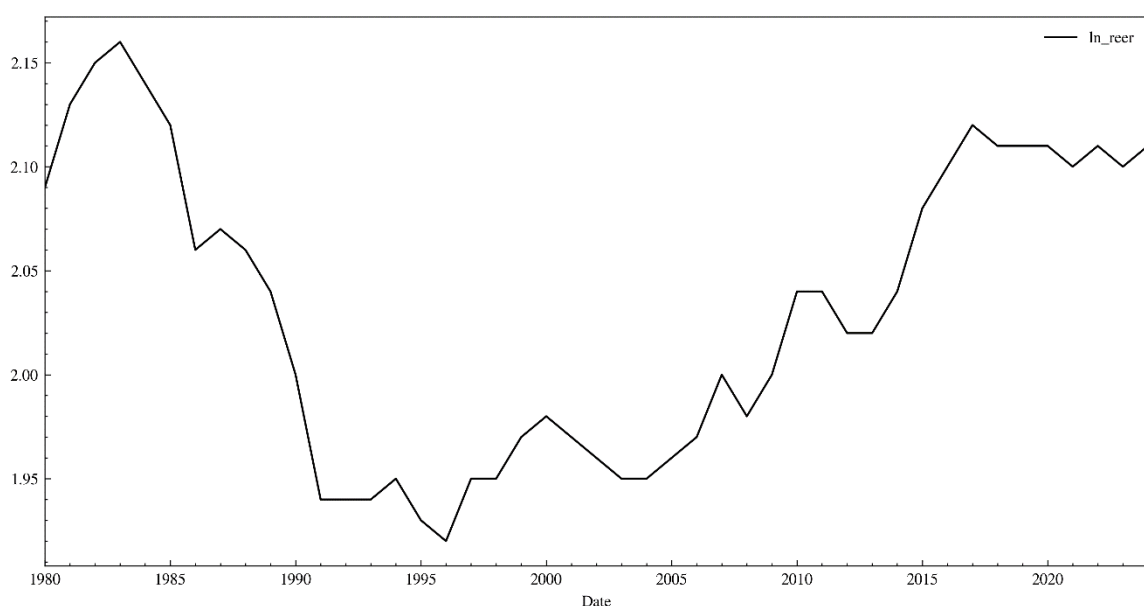
After the point in the 1990s the REER started to go up again but in a choppy way. It went up and down a bit. This showed that the economy was changing. It was adjusting after trade reforms the financial crisis at the end of the 1990s and changes in global inflation.

Starting around 2010 the REER started to go up. It stayed high in the 2020s. This shows that the Nepalese Rupee became much stronger in terms against other currencies. Since the NPR is basically tied to the Rupee (INR) this strong movement against other currencies like the USD and the Euro is because of two reasons. One is that inflation in Nepal was higher than in countries. The other is that the INR and the NPR got stronger compared to currencies during this time.

The strong movement of the REER after 2010 helps explain the growing trade deficit in Figure 2. As the REER went up exports, from Nepal became more expensive in the world. At the time imports became cheaper in Nepal. This situation made the trade balance with the rest of the world worse.

Figure 5

Real effective exchange rate in log form



Source: Bruegel database

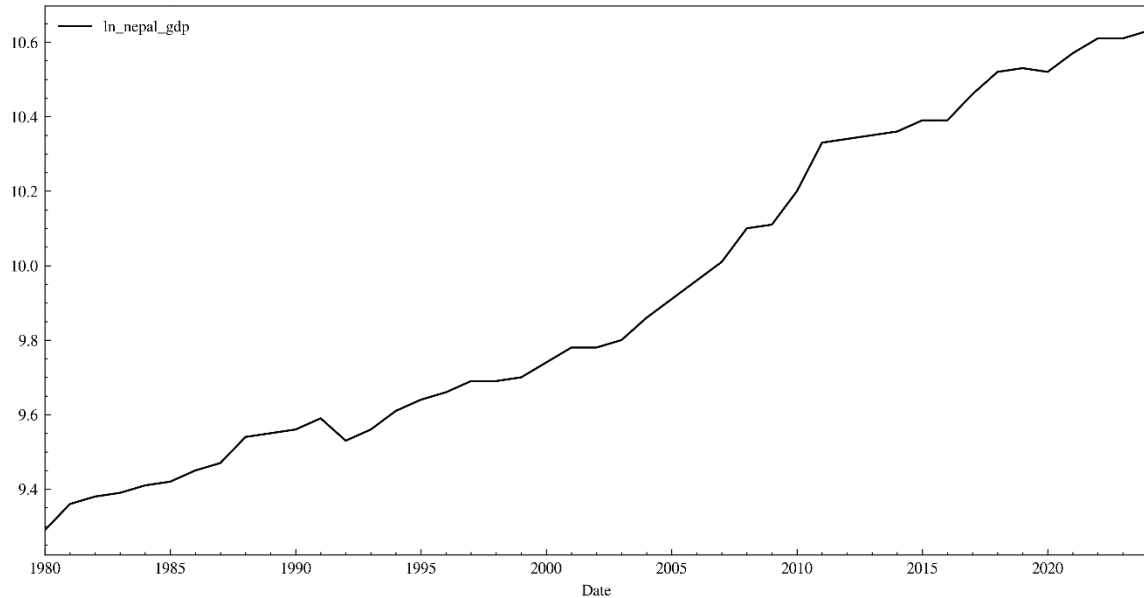
4.1.2 Trend of Nepal GDP

Real GDP is used as a control variable to capture the effects of absorption. When domestic income goes up import demand usually rises well. Figure 6 shows that Nepal's real GDP has followed an upward path from 1980 to 2024. The trend looks smooth overall with dips during times of crisis—like the 2008 financial crisis, the 2015 earthquake and the 2020 COVID-19 pandemic. These events caused slowdowns in economic growth but the long-term pattern remains one of expansion. As domestic income keeps growing so

does the need for imports. Nepal continues to depend on imported capital goods, raw materials and petroleum. This dependence means that every gain in income pushes import demand higher. That in turn puts downward pressure on the trade balance. This effect happens without any changes, in exchange rates.

Figure 6

Gross domestic product of Nepal in log form



Source: WDI database

4.1.3 Transition to Inferencing

The visual trends show a puzzle that needs solving: even though the currency has gotten weaker over time and the economy has grown a little the trade balance in Nepal has kept getting worse. Also the different ways the trade balance works with India and other countries outside of Nepal suggests that something might be wrong, with how the data's put together. To check these ideas and find out if the trade balance isn't improving because of effects or because the model is not right we now move to the statistical analysis. We start by checking if the data stays the same over time i.e. stationarity tests.

4.2 Pre-Estimation Diagnostics: Unit Root Tests

Before estimating the ARDL and NARDL models, it is essential to make sure that no variable is integrated of order two i.e. $I(2)$ because the bounds testing approach becomes invalid when $I(2)$ variables are present. To verify this Augmented Dickey-Fuller (ADF)

and Phillips-Perron (PP) unit root tests to every log-transformed variable checking both the level and the first difference.

Table 2

Unit Root Test Results

Variable	ADF (Level)	PP (Level)	ADF (1st Diff.)	PP (1st Diff.)	Decision
ln total tbr	-1.448 (0.549)	-1.151 (0.686)	-4.476 (0.000)***	-4.321 (0.001)***	I(1)
ln reer	-1.620 (0.463)	-1.141 (0.690)	-4.817 (0.000)***	-4.813 (0.000)***	I(1)
ln nepal gdp	0.386 (0.980)	0.379 (0.979)	-5.917 (0.000)***	-5.904 (0.000)***	I(1)
ln india tbr	-1.690 (0.429)	-1.884 (0.336)	-5.593 (0.000)***	-5.723 (0.000)***	I(1)
ln row tbr	-1.662 (0.443)	-1.585 (0.481)	-6.121 (0.000)***	-6.503 (0.000)***	I(1)

Note. Values in parentheses represent p-values. *** indicates significance at the 1% level.

From Table 2 the null hypothesis that any variable contains a unit root cannot be rejected at the level of any variable. This result matches the intuition drawn from the trend plots, which show that the variables possess stochastic trends. When the first difference is taken every variable becomes stationary at the one percent significance level in both the ADF and PP tests. Therefore each variable is integrated of order one, I(1). This fact satisfies the prerequisite, for applying both the linear ARDL and NARDL bounds-testing approaches.

4.3 Stage 1: Linear ARDL Baseline Estimation

The first step of the analysis is to estimate a simple ARDL model so that we have a symmetric baseline. The best lag length was chosen automatically using the Schwarz Information Criterion (SIC). We check whether a long-run cointegrating relationship exists by using the Pesaran et al. (2001) F-bounds test. The F-bounds test shows a split between the aggregated or pegged trade balance and the floating regime trade balance. For the model and the India-specific model, the F-statistics (2.009 and 0.962 respectively) are below the lower bound I (0) at the 5% significance level. Therefore we cannot reject the hypothesis of no cointegration, which means there is no stable long-run equilibrium between the trade balance, REER and GDP in these two regimes. On the hand the ROW model gives an F-statistic of 5.731, which is higher than the upper bound I (1) of 3.87 at the 5% level

confirming that cointegration is present. The long-run estimates, for the ROW model reveal that the REER coefficient is -1.767 and statistically significant while domestic GDP has an significant effect (-0.281). I notice that this supports the observation that rising domestic income expands the trade balance by boosting import demand.

Table 3

Linear ARDL Bound Test, Long-Run Coefficients, and Model Fit

Model	Selected ARDL	F-Statistic	5% Lower Bound	5% Upper Bound	Cointegration?	R²	Adj. R²
Aggregate	(2, 0, 0)	2.009	3.100	3.870	No	0.919	0.911
India	(1, 0, 0)	0.962	3.100	3.870	No	0.836	0.824
ROW	(2, 0, 0)	5.731	3.100	3.870	Yes	0.826	0.808

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ln reer	-1.767	0.291	-6.059	0.000***
ln nepal gdp	-0.281	0.049	-5.632	0.000***
C	5.745	0.609	9.421	0.000***

Note. The critical bounds values are based on the 5% significance level. *** indicates statistical significance at the 1% level.

4.4 Stage 2: Error Correction Model (ECM) Estimation

On the bounds test results the Error Correction Model or ECM was derived to capture the short-run dynamics and the speed of adjustment back to the long-run equilibrium. The lagged error correction term or ECT was extracted from the conditional error correction regression. For the ROW model the ECT coefficient is -0.839 which is highly significant and correctly signed. This indicates a speed of convergence to the long-run equilibrium adjusting approximately 83.9% of the disequilibrium in each period. For the model and the India model despite the lack of cointegration established by the

F-bounds test the ECT terms are negative and weakly significant. This anomaly is common in samples with $T = 43$, where the t-test on the ECT may possess more power than the F-bounds test as noted by (Kremers et al., 1992). However strictly adhering to the bounds testing protocol the J-Curve interpretation is only fully applicable, to the ROW regime.

Table 4

Short-Run and Long-Run Adjustment Dynamics

Model	Short Run REER		Error Correction Term		
	Coefficient	Prob.	ECT (-1)	t-statistics	Prob.
Aggregate	-0.382	0.163	-0.335	-2.685	0.010***
India	-0.145	0.557	-0.150	-1.755	0.086*
ROW	-1.484	0.002***	-0.839	-4.572	0.000***

Note. ECT = Error Correction Term. REER = Real Effective Exchange Rate. * $p < .10$, ** $p < .05$, *** $p < .01$

4.5 Stage 3: Non-linear ARDL (NARDL) Estimation

The third stage of the empirical analysis moves from the model to the asymmetric model by breaking down the Real Effective Exchange Rate (REER) into two parts: a positive part that shows real currency going up in value and a negative part that shows real currency going down in value. This splitting is the part of the Non-linear Autoregressive Distributed Lag (NARDL) method created by Shin et al. (2014). It helps the model find answers for when the currency goes up and when it goes down instead of using one answer for both. The goal of this step is to check if there are effects—meaning if real currency going up and real currency going down affect the trade balance in different ways for the whole country, for India and for the Rest of the World. When setting up the NARDL models, the best number of lags was chosen using information criteria. In all three groups the criteria picked a structure for both the positive and negative parts. This means only the current values of the exchange rate are used in the regression. This result comes from the way the data chooses the lags and is not because the model is limited. It also makes sense with the idea of short-term symmetry from the peg. This idea says that Nepal's real peg to the Rupee stops the short-term changes in real exchange rate from affecting the trade balance. It does this by managing shocks through reserves, controlled actions and rules on

money flow of letting the shocks change prices and trade. In this situation the immediate effect of a real exchange rate shock should be more important, than any changes later. The choice of lag-0 for the parts fits with both the way the model works and how the Nepalese economy behaves.

The NARDL bounds test confirms the earlier ARDL results. In this case and in the India case the test shows that cointegration is missing. In the ROW case however the test shows a cointegration ($F = 8.318 > 3.67$). In the ROW case the long-run asymmetric numbers give us clues. The number for depreciation called ($REER_{NEG}$) is -1.800 and it is very significant. The number for appreciation called ($REER_{POS}$) is basically zero (0.245, $p = 0.774$). This means that when the NPR falls in value against currencies around the world Nepal's trade balance, with the ROW gets clearly worse. In the case ($REER_{NEG}$) is only barely significant, which hints that the same pattern may exist at the national level.

Table 5

NARDL Bounds Test, Long-Run Asymmetric Coefficients, and Model Fit

Model Selected	NARDL Order	F-Statistic	5% Bounds	Cointegration?	R²	Adj. R²
Aggregate	(2, 0, 0, 0)	1.711	[2.79, 3.67]	No	0.92	0.91
India	(1, 0, 0, 0)	0.979	[2.79, 3.67]	No	0.841	0.824
ROW	(3, 0, 0, 0)	8.318	[2.79, 3.67]	Yes	0.898	0.881

Regime	Variable	Coefficient	Std. Error	t-Statistic	Prob.	Significance
Aggregate	REER POS	0.666	2.588	0.257	0.798	*
	REER NEG	-1.541	0.765	-2.014	0.051	
	GDP	-0.989	0.772	-1.280	0.208	
India	REER POS	6.749	9.391	0.718	0.476	
	REER NEG	-2.656	2.912	-0.911	0.367	
	GDP	-2.606	2.867	-0.909	0.368	
ROW	REER POS	0.245	0.849	0.288	0.774	***
	REER NEG	-1.800	0.331	-5.436	0.000	
	GDP	-0.848	0.257	-3.299	0.002	

Note. REER POS and REER NEG represent positive and negative changes in the real effective exchange rate, respectively. Critical bounds values are based on the 5% significance level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

4.5.1 Asymmetry Testing (Wald Test)

To formally test Hypothesis 2, the long-run Wald restriction $\theta^+ = \theta^-$ was evaluated.

Table 6

Long-Run Wald Test for Asymmetry

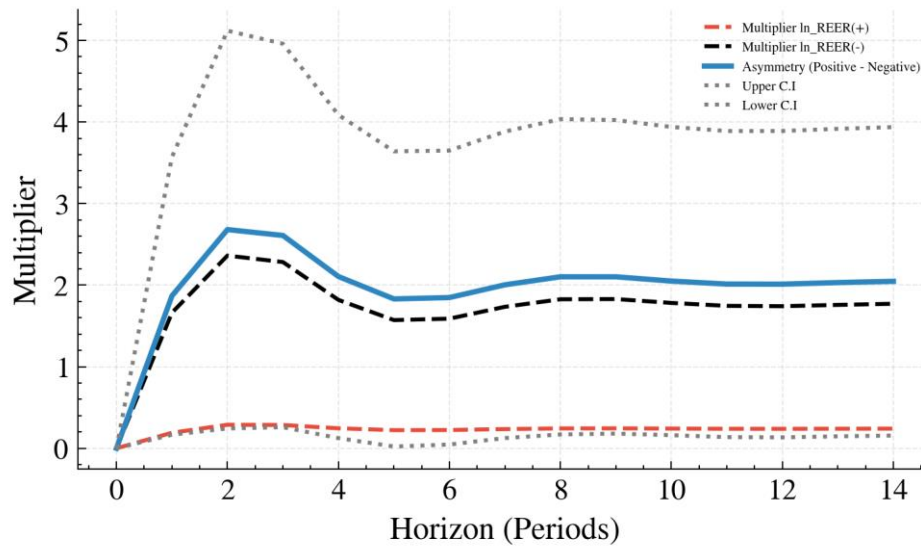
Model	Wald F-statistics	Prob.	Decision (5% level)
Aggregate	0.604	0.441	Symmetric
India	1.041	0.313	Symmetric
ROW	3.467	0.071	Asymmetric (at 10%)

Note. The Wald test examines the null hypothesis of long-run symmetry between positive and negative changes in the real effective exchange rate. Rejection of the null hypothesis indicates asymmetric long-run effects

4.5.2 NARDL ROW Dynamic Multipliers

Figure 7

NARDL rest of world model dynamic multipliers



Source: calculation based on appendix 2

Figure 7 shows the multipliers for the ROW model. The depreciation multiplier goes up quickly and then stays around 1.8 – this number is the same as the long run coefficients ($\theta^- = -1.8$) which shows that real depreciation makes the trade balance worse forever with no improvement. On the hand the appreciation multiplier stays close to zero which matches the small long run coefficient($\theta^+ = 0.245$). The difference between the two lines shows that there is an asymmetry in the run. Both multipliers finish adjusting in two periods. The fact that the depreciation line does not go down and then come back

up proves visually that the J-Curve is not present in Nepal's ROW trade. Instead depreciation makes the deficit worse, over time – this is called the "Worsening Curve".

4.6 Diagnostic Testing

To ensure the correctness of the NARDL models many diagnostic tests were carried out including the Breusch-Godfrey (BG) test for serial correlation the Breusch-Pagan-Godfrey (BPG) test for unequal variance and the Ramsey RESET test, for functional form.

Table 7

NARDL Diagnostic Tests (p-value)

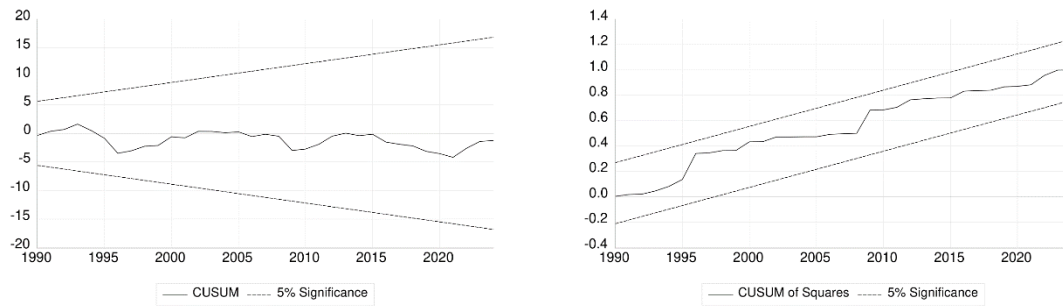
Model	BG Test (Serial Corr.)	BPG Test (Heterosced.)	Ramsey RESET (Form)
Aggregate	0.991	0.022**	0.515
India	0.883	0.736	0.586
ROW	0.730	0.509	0.644

Note. BG denotes the Breusch–Godfrey test for serial correlation, BPG denotes the Breusch–Pagan–Godfrey test for heteroscedasticity, and Ramsey RESET tests model specification. Values reported are p-values. ** indicates statistical significance at the 5% level.

All models are free from correlation and functional form misspecification according to the results. India models and ROW models show homoscedasticity. The aggregate model shows signs of heteroscedasticity ($p=0.022$). The heteroscedasticity is not unusual in small-sample macroeconomic time series that include shocks, such as the 2015 earthquake and the COVID-19 pandemic. Still this heteroscedasticity does not break the cointegration inference. To save space the CUSUM and CUSUMSQ plots for the ROW NARDL model—the model that shows strong long-run cointegration and asymmetry—are in Figures 8. The CUSUM statistics and CUSUMSQ statistics stay inside the 5% bounds proving that the parameters are stable over the sample period even, with the exogenous shocks.

Figure 8

CUSUM and CUSUMSQ



Source: author's calculations using Eviews

4.7 Hypothesis Testing Summary

Based on the sequential estimations, the three research hypotheses are resolved as follows:

Hypothesis 1 (J-Curve Effect): **Rejected.**

A J-Curve requires a short-run deterioration followed by a long-run improvement (positive long-run coefficient for depreciation) conditional on cointegration. Cointegration was only established for the ROW regime. In the ROW NARDL model the run coefficient for real depreciation ($REER_{NEG}$) is -1.800 (negative and highly significant). Because depreciation worsens the trade balance in the run rather than improving it the J-Curve phenomenon does not exist for Nepal. This finding aligns with the view that Nepal's heavy reliance on inelastic essential imports prevents the volume effects of depreciation from outweighing the price effects.

Hypothesis 2 (Asymmetric Adjustment): **Partially Supported.**

The null of long-run symmetry is rejected for the ROW model at the 10% significance level (Wald $F = 3.467$, $p = 0.071$). The aggregate and India models exhibit symmetry. Therefore asymmetry is a regime- phenomenon rather than a universal one across Nepal's trade data.

Hypothesis 3 (Regime-Dependent Asymmetry): **Supported.**

The empirical evidence strongly favors regime dependency. The India (pegged) model demonstrated no cointegration and symmetric adjustment confirming that the bilateral peg effectively mutes the transmission of exchange rate shocks to trade flows.

Conversely the ROW (floating) model displayed cointegration and significant long-run asymmetry. While the linear ECM estimated a convergence speed of -0.839 for the ROW model, re-estimating the model within the NARDL-ECM framework reveals a faster adjustment speed ($ECT = -0.926$). This indicates that allowing for asymmetry improves the model's speed of correction to run equilibrium. Aggregating these two distinct regimes masks the dynamics of the floating market proving that exchange rate policy is only relevant for Nepal's trade, with the Rest of the World.

4.8 Discussion of Findings

The empirical results of this study offer an explanation for the long-standing puzzle in Nepal's trade deficit and exchange rate behavior. For years traditional economic theory predicted that the continuous depreciation of the Nepalese Rupee should help reduce the trade deficit. Yet this did not happen. This failure has been a mystery in Nepal's macroeconomic analysis.

The study shows that the J-Curve effect, which usually helps improve trade balances after currency depreciation is not working in Nepal. This is not because the model is wrong or misspecified but rather due to deep structural issues in the economy. When we look at the data through a disaggregated regime approach these structural constraints become clear.

First the analysis of trade with India supports the idea that the pegged exchange rate system is the key. Since the Nepalese Rupee is tied to the Rupee real exchange rate changes mostly come from inflation differences. The central bank manages reserves to absorb these changes, which means the exchange rate does not adjust in a way that affects trade. The NARDL bounds test showed no relationship and the Wald test confirmed symmetric responses. This means that changes in currency value do not affect the trade balance in either direction—because the peg removes the price mechanism that would normally drive trade adjustments. This finding extends the work of Adhikari (2020) showing that in a pegged system trade flows react symmetrically—or not at all—to changes, in currency value.

Second, the analysis of the Rest of the World (ROW) trade reveals where the true asymmetric dynamics lie. When the REER is allowed to decompose into appreciation and depreciation, the long-run Wald test rejects symmetry at the 10% level. Specifically, the

coefficient for real depreciation (-1.800) is highly significant, while the coefficient for appreciation (0.245) is insignificant. This indicates an inherent structural asymmetry: currency depreciation systematically worsens the trade balance, but currency appreciation does not correspondingly improve it. The significantly negative long-run coefficient for real depreciation provides empirical confirmation that the Marshall-Lerner condition does not hold for Nepal's trade with the rest of the world. Because the sum of import and export price elasticities is less than one, the volume effect of depreciation is entirely dominated by the adverse price effect of more expensive essential imports. As Pathak (2020) noted, when demand for essential imports like petroleum, machinery, and pharmaceuticals is price-inelastic, a depreciation simply raises the domestic NPR cost of survival without reducing import volumes. Meanwhile, Nepalese exporters lack the productive capacity to immediately capitalize on the cheaper currency.

Third the J-Curve did not appear over the term (Hypothesis 1) which supports earlier worries of Chaulagai (2015) who found an "L-Curve" using linear models. This study used an advanced NARDL framework to check if non-linearity was hiding the J-Curve. The NARDL long-run coefficients for the ROW regime show that depreciation has a lasting effect on the trade balance. So of an initial worsening followed by improvement (J-Curve) or a flat response (L-Curve as previous linear studies said) Nepal's trade balance with the ROW shows a "Worsening Curve". Where depreciation makes the deficit worse over time because of deep import dependence. The "L-curve" finding was a mistake from using linear models. They forced symmetry. Averaged out the different effects making a worsening trade balance look like just no change. Currency depreciation, against the USD and other global currencies acts like an inflation tax of helping exports.

Finally the results strongly support the decision to break down the trade data. If this study had only used data the F-bounds test would not have found any long-term connection. The important asymmetric effects happening in the ROW regime would have been hidden because of the peg. The disaggregated NARDL method correctly separated the floating-regime effects showing that exchange rate policy alone is not a way to fix Nepal's trade deficit.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of the Study

The impact of exchange rate fluctuations on trade balance has been a subject of debate for a long time now. Nepal has been experiencing a perennial and expanding trade deficit that remains unfixed despite all conventional measures. As per the Marshall-Lerner condition, trade balance should improve through depreciation of local currency against foreign currencies. However, persistent depreciation of Nepalese Rupee (NPR) against convertible currencies over the last two decades has not yielded the intended result. Several researchers have attempted to explore the reason for such an occurrence, but their linear analysis failed to present any substantial conclusion. The existing literature on the topic has worked on a symmetric foundation where response of trade balance to appreciation and depreciation of currency was assumed to be same. Therefore, concluding Nepal's trade balance to suffer from L-curve with no response to change in exchange rates.

This thesis attempted to explore the issue with an application of non-linear autoregressive distributed lag (NARDL) approach, developed by Shin et al. (2014), to unveil asymmetric J-curve in case of Nepal. Analysis was performed using annual time series data covering 45 years (1980-2024). Trade balance of Nepal has been divided into three different regimes; namely, aggregate trade balance, bilateral trade balance with India (maintaining fixed exchange rate), and trade balance with rest of the world (maintaining floating exchange rates). Three different stages of analysis; linear ARDL bounds testing, Error Correction Model and NARDL model, were performed on each regime. The aim of the study was to re-examine whether the reason for not finding J-curve as found in earlier literature was because of methodological inconsistency or some other reason.

5.2 Major Findings

The analysis revealed three main results related to testing hypotheses and answering the posed research question. Firstly, the study did not find a J-Curve phenomenon for any of the three trade regimes, which rejects Hypothesis 1 and resolves the paradox. The F-

bounds test for the aggregate and India models showed that there is no cointegration in the long run, which renders the J-curve irrelevant for these two cases. For the third model (ROW), the test confirmed long-run cointegration, but the short- and long-run coefficients for real depreciation were significantly negative (-1.484 and -1.800). As a result, depreciation worsens the trade balance in the short and long run, and there is no J-curve for Nepal.

Secondly, the analysis found that long-run asymmetric adjustment is present but is regime-dependent, which confirms Hypothesis 2. The NARDL Wald test showed that for the aggregate and India models, asymmetry is not inherent in the long-run adjustment process. However, for the ROW model, the test revealed a significant long-run asymmetry (Wald $F = 3.467$, $p = 0.071$). The analysis found opposite long-run effects of appreciation and depreciation for the ROW model. Results show that real depreciation worsens the trade balance by 1.8%, while real appreciation does not affect it (0.245 ; $p = 0.774$). The study confirmed the presence of a significant long-run asymmetry for one of the three trade regimes.

Thirdly, the study confirmed Hypothesis 3 and showed that analyzing the trade balance under different exchange rate regimes is essential. The India (pegged) model did not exhibit long-run relationship (no cointegration) and long-run asymmetry, while the ROW (floating) model did. In other words, the pegged exchange rate regime dampened the effect of exchange rate fluctuations on trade flows. The analysis of aggregate trade flows misrepresents the dynamics of trade balance under floating exchange rates because the aggregate results conceal significant differences in trade balance adjustment between the two exchange rate regimes. Specifically, the ROW model demonstrates a high speed of adjustment (-0.926 in NARDL-ECM) and a significant long-run asymmetry.

5.3 Conclusions

These findings offer a fascinating resolution to the paradox of the trade deficit in Nepal. It was not just methodological misspecification but structural nuances that manifested through disaggregated data

Firstly, the analysis of bilateral India trade affirmed theoretical foundations of the pegged exchange rate regime. As the NPR was pegged to the INR, real exchange rate variations were more attributable to inflation differentials which were smoothed by central

bank reserve management rather than being manifested in price ratios. In essence, the exchange rate mechanism failed as a short- or long-run corrective mechanism for the India trade balance. Structural determinants dictated trading flows with India comprising of supply-demand factors, agricultural produce, and industrial capacities.

Further, the analysis of ROW trade exposed the central structural aberration. Notably, the substantially negative long-run coefficient of real depreciation found empirical validation in the Marshall-Lerner theorem which posits that the sum of the import and export price elasticities must be greater than one for depreciation to improve the trade balance. This finding suggests that the depreciation of the NPR did not lead to an improvement in the trade balance with the ROW. The price mechanism of traded volumes was overwhelmed by the adverse price impact of depreciation on vital imports. In other words, the price elasticity of demand for crucially inelastic imports outweighed the income elasticity of demand for exports. Precisely, because Nepal trades in inelastic imports such as petroleum and pharmaceuticals and has no competitive export capacities, depreciation simply raised the NPR cost of living without improving export volumes.

By and large, these findings suggest that the trade deficit in Nepal cannot be resolved through exchange rate interventions alone because the country's exchange rate regime is structurally compromised. Further, the analysis proved that the application of a linear specification was misleading and produced an "L-Curve" that misrepresented the true nature of the trade dynamics. In essence, the conventional J-curve analysis of the relationship between currency depreciation and trade deficits was flawed in the case of Nepal because it resulted in a "Worsening Curve" that showed a worsening trade deficit with consecutive depreciations of the NPR. Specifically, there was a simultaneous adverse effect of depreciation on the trade balance, first in the short run and then in the long run, unlike the conventional J-Curve which predicts an initial adverse impact followed by an improvement. This was primarily because currency devaluation against hard currencies was an inflationary measure that had little impact on export performance. The "L-Curve" was a methodological flaw that emanated from applying a linear specification. The findings proved that the use of pooled data to analyze trade deficits and surpluses is flawed and produces misleading results.

5.4 Policy Recommendations

Based on the findings, the following policy recommendations are made to the Government of Nepal, Nepal Rastra Bank (NRB) and the economic ministries:

Abolishing depreciation as an avenue to correct the deficit: The policy makers should acknowledge that depreciating the NPR will not help in correcting the trade deficit since import demand is inelastic and the country has limited export capacity. Thus, depreciation will only lead to higher imported inflation, higher cost of living and an expanded current account deficit.

Promoting structural import substitution and diversification of exports: The government should focus more on promoting structural import substitution and diversification of exports. Since price competitiveness through depreciation has proven to be a failure, Nepal needs to focus on non-price competitiveness. Policies should be implemented to encourage production of goods that are available in import, especially agricultural products and consumer goods. In addition, export diversification should be emphasized by shifting focus from primary goods to manufactured goods. This will ensure that Nepal has competitive products in the international market.

Keeping the India peg for stability: The analysis reveals that the peg with India has insulated bilateral trade from the volatility in the exchange rates and smoothened shocks. In addition, since India accounts for the largest share of imports, especially petroleum products, it is advisable for Nepal to keep the peg with India for macroeconomic stability. However, instead of currency devaluation, the government should focus on promoting competitive products in the Indian market.

Controlling inflation to avoid real appreciation: The study established that real depreciation after 2010 of NPR against the ROW has hurt the country's export competitiveness. However, since the nominal exchange rate against USD is floating, the only way to prevent real appreciation is to control the rate of domestic inflation. The bank should control money supply to avoid a situation where inflation rate in Nepal is much higher than that of the trading partners.

Encouraging remittance-based capital: Since Nepal has been receiving remittance amounting to almost 25% of the GDP, which ends up being used in importing goods and services, there is a need for the government to develop fiscal policies that will encourage

alternative uses of remittance. The policies will ensure that the huge amount of cash being sent to Nepal is channeled to productive uses, rather than the current situation where most of it ends up in the real estate sector.

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APPENDIX 1

RAW DATA

exports_india_npr_mn, exports_row_npr_mn, exports_total_npr_mn, gdp_npr_mn, imports_india_npr_mn,
imports_total_npr_mn, imports_row_npr_mn are in millions

year is in per annum

reer is the real effective exchange rate

exports_india_npr_mn	exports_row_npr_mn	exports_total_npr_mn	gdp_npr_mn	imports_india_npr_mn	imports_total_npr_mn	imports_row_npr_mn	reer	year
992.40	616.30	1,608.70	6.2E+10	2,179.00	4,428.20	2,249.20	124.26	1980
994.40	497.10	1,491.50	7.3E+10	2,280.90	4,930.30	2,649.40	135.89	1981
843.30	288.70	1,132.00	8.1E+10	2,499.60	6,314.00	3,814.40	141.19	1982
1,160.70	543.20	1,703.90	8.8E+10	3,058.00	6,514.30	3,456.30	145.04	1983
1,601.70	1,138.90	2,740.60	1.1E+11	3,895.80	7,742.10	3,846.30	139.31	1984
1,241.10	1,836.90	3,078.10	1.3E+11	3,970.90	9,341.20	5,370.30	131.51	1985
1,302.60	1,688.80	2,991.40	1.5E+11	4,262.00	10,905.20	6,643.20	116.05	1986
1,567.60	2,546.90	4,114.50	1.6E+11	4,595.70	13,869.60	9,273.90	116.37	1987
1,034.90	3,160.40	4,195.30	1.9E+11	4,238.70	16,263.70	12,025.00	115.41	1988
602.50	4,553.70	5,156.20	2.3E+11	4,674.50	18,324.90	13,650.40	108.84	1989
1,552.20	5,835.30	7,387.50	2.6E+11	7,323.10	23,226.50	15,903.40	100.81	1990
1,450.00	12,256.50	13,706.50	3.7E+11	11,245.50	31,940.00	20,694.50	86.20	1991
1,621.70	15,644.80	17,266.50	4.2E+11	12,542.10	39,205.60	26,663.50	87.89	1992
2,408.90	16,884.50	19,293.40	4.7E+11	17,035.40	51,570.80	34,535.40	86.81	1993
3,124.30	14,514.90	17,639.20	5.1E+11	19,615.90	63,679.50	44,063.60	88.71	1994
3,682.60	16,198.50	19,881.10	5.9E+11	24,398.60	74,454.50	50,055.90	84.62	1995

5,226.20	17,410.30	22,636.50	6.4E+11	24,853.30	93,553.40	68,700.10	82.82	1996
8,794.40	18,719.10	27,513.50	7.3E+11	27,331.00	89,002.00	61,671.00	89.79	1997
12,530.70	23,145.60	35,676.30	8.2E+11	32,119.70	87,525.30	55,405.60	88.60	1998
21,220.70	28,602.00	49,822.70	8.7E+11	39,660.10	108,504.90	68,844.80	94.30	1999
26,030.20	29,623.90	55,654.10	9.9E+11	54,700.90	115,687.20	60,986.30	95.86	2000
27,956.20	18,988.60	46,944.80	1.1E+12	56,622.10	107,388.90	50,766.90	94.39	2001
26,430.00	23,500.60	49,930.60	1.1E+12	70,924.20	124,352.10	53,427.90	91.00	2002
30,777.10	23,133.60	53,910.70	1.1E+12	78,739.50	136,277.10	57,537.60	88.52	2003
38,916.90	19,788.80	58,705.70	1.1E+12	88,675.50	149,473.60	60,798.10	88.64	2004
40,714.70	19,519.40	60,234.10	1.1E+12	107,143.10	173,780.30	66,637.20	91.88	2005
41,728.80	17,654.30	59,383.10	1.2E+12	115,872.30	194,694.60	78,822.30	93.97	2006
38,555.70	20,710.80	59,266.50	1.1E+12	142,376.50	221,937.80	79,561.30	100.00	2007
41,005.90	26,691.60	67,697.50	1.4E+12	162,437.60	284,469.60	122,032.00	95.64	2008
39,993.70	20,830.30	60,824.00	1.4E+12	217,114.30	374,335.20	157,220.90	99.33	2009
43,360.40	20,978.10	64,338.50	1.4E+12	261,925.20	396,175.50	134,250.30	108.60	2010
49,616.30	24,644.70	74,261.00	1.7E+12	299,389.60	461,667.70	162,278.10	110.15	2011
51,000.01	23,831.59	76,917.06	1.9E+12	367,031.19	556,740.24	127,257.68	105.36	2012
59,613.71	29,537.04	91,991.41	2.2E+12	477,947.04	714,365.85	163,100.21	105.38	2013
55,864.57	27,224.56	85,319.08	2.3E+12	491,655.91	774,684.10	182,861.89	108.96	2014
39,493.74	28,941.90	70,117.17	2.6E+12	477,212.59	773,599.15	180,692.14	120.05	2015
41,449.21	29,898.37	73,049.14	2.6E+12	633,669.61	990,113.20	229,198.62	125.88	2016
46,719.76	32,202.30	81,359.80	2.8E+12	814,101.62	1,245,103.22	271,014.52	130.68	2017
62,731.84	32,267.88	97,109.52	3.2E+12	917,922.21	1,418,535.34	295,094.49	127.55	2018
70,108.89	26,409.04	97,709.11	3.6E+12	735,294.82	1,196,799.06	279,583.93	129.52	2019
106,372.05	33,735.97	141,124.08	3.5E+12	971,603.95	1,539,837.07	334,310.06	129.44	2020
155,222.30	43,999.90	200,030.96	3.8E+12	1,200,152.73	1,920,448.35	455,511.91	127.29	2021
106,686.40	48,688.50	157,140.70	4.3E+12	1,027,847.54	1,611,731.77	361,168.30	127.77	2022
103,177.05	46,615.14	152,380.80	4.5E+12	996,682.16	1,592,985.53	297,528.72	125.67	2023
224,684.40	49,715.78	277,030.20	4.8E+12	1,071,198.94	1,804,122.73	391,819.20	127.83	2024

Source: Nepal Rastra Bank, World Bank Database, Bruegel database

APPENDIX 2

IN NATURAL LOG FORM

ln_export _india	ln_export _row	ln_im port_ india	ln_india _tbr	ln_impor t_row	ln_india _gdp	ln_nepal _gdp	ln_reer	ln_reer_ neg	ln_row _gdp	ln_reer_ pos	ln_row_ tbr	ln_total _exports	ln_total _tbr	ln_total _import	year
3.00	2.79	3.34	(0.34)	3.35	11.27	9.29	2.09		13.94		(0.56)	3.21	(0.44)	3.65	1980
3.00	2.70	3.36	(0.36)	3.42	11.29	9.36	2.13	-	13.95	0.04	(0.73)	3.17	(0.52)	3.69	1981
2.93	2.46	3.40	(0.47)	3.58	11.30	9.38	2.15	-	13.95	0.06	(1.12)	3.05	(0.75)	3.80	1982
3.06	2.73	3.49	(0.42)	3.54	11.34	9.39	2.16	-	13.95	0.07	(0.80)	3.23	(0.58)	3.81	1983
3.20	3.06	3.59	(0.39)	3.59	11.33	9.41	2.14	(0.02)	13.96	0.07	(0.53)	3.44	(0.45)	3.89	1984
3.09	3.26	3.60	(0.51)	3.73	11.37	9.42	2.12	(0.04)	13.98	0.07	(0.47)	3.49	(0.48)	3.97	1985
3.11	3.23	3.63	(0.51)	3.82	11.40	9.45	2.06	(0.10)	14.05	0.07	(0.59)	3.48	(0.56)	4.04	1986
3.20	3.41	3.66	(0.47)	3.97	11.45	9.47	2.07	(0.10)	14.10	0.08	(0.56)	3.61	(0.53)	4.14	1987
3.01	3.50	3.63	(0.61)	4.08	11.47	9.54	2.06	(0.11)	14.15	0.08	(0.58)	3.62	(0.59)	4.21	1988
2.78	3.66	3.67	(0.89)	4.14	11.47	9.55	2.04	(0.13)	14.17	0.08	(0.48)	3.71	(0.55)	4.26	1989
3.19	3.77	3.86	(0.67)	4.20	11.51	9.56	2.00	(0.17)	14.23	0.08	(0.44)	3.87	(0.50)	4.37	1990
3.16	4.09	4.05	(0.89)	4.32	11.43	9.59	1.94	(0.23)	14.24	0.08	(0.23)	4.14	(0.37)	4.50	1991
3.21	4.19	4.10	(0.89)	4.43	11.46	9.53	1.94	(0.23)	14.27	0.08	(0.23)	4.24	(0.36)	4.59	1992
3.38	4.23	4.23	(0.85)	4.54	11.45	9.56	1.94	(0.23)	14.28	0.08	(0.31)	4.29	(0.43)	4.71	1993
3.49	4.16	4.29	(0.80)	4.64	11.51	9.61	1.95	(0.23)	14.31	0.09	(0.48)	4.25	(0.56)	4.80	1994
3.57	4.21	4.39	(0.82)	4.70	11.56	9.64	1.93	(0.25)	14.36	0.09	(0.49)	4.30	(0.57)	4.87	1995
3.72	4.24	4.40	(0.68)	4.84	11.59	9.66	1.92	(0.26)	14.37	0.09	(0.60)	4.35	(0.62)	4.97	1996
3.94	4.27	4.44	(0.49)	4.79	11.62	9.69	1.95	(0.26)	14.37	0.12	(0.52)	4.44	(0.51)	4.95	1997
4.10	4.36	4.51	(0.41)	4.74	11.62	9.69	1.95	(0.26)	14.37	0.12	(0.38)	4.55	(0.39)	4.94	1998
4.33	4.46	4.60	(0.27)	4.84	11.66	9.70	1.97	(0.26)	14.38	0.14	(0.38)	4.70	(0.34)	5.04	1999
4.42	4.47	4.74	(0.32)	4.79	11.67	9.74	1.98	(0.26)	14.40	0.15	(0.31)	4.75	(0.32)	5.06	2000
4.45	4.28	4.75	(0.31)	4.71	11.69	9.78	1.97	(0.27)	14.39	0.15	(0.43)	4.67	(0.36)	5.03	2001
4.42	4.37	4.85	(0.43)	4.73	11.71	9.78	1.96	(0.28)	14.41	0.15	(0.36)	4.70	(0.40)	5.09	2002

4.49	4.36	4.90	(0.41)	4.76	11.78	9.80	1.95	(0.29)	14.46	0.15	(0.40)	4.73	(0.40)	5.13	2003
4.59	4.30	4.95	(0.36)	4.78	11.85	9.86	1.95	(0.29)	14.52	0.15	(0.49)	4.77	(0.41)	5.17	2004
4.61	4.29	5.03	(0.42)	4.82	11.91	9.91	1.96	(0.29)	14.56	0.16	(0.53)	4.78	(0.46)	5.24	2005
4.62	4.25	5.06	(0.44)	4.90	11.97	9.96	1.97	(0.29)	14.60	0.17	(0.65)	4.77	(0.52)	5.29	2006
4.59	4.32	5.15	(0.57)	4.90	12.09	10.01	2.00	(0.29)	14.66	0.20	(0.58)	4.77	(0.57)	5.35	2007
4.61	4.43	5.21	(0.60)	5.09	12.08	10.10	1.98	(0.31)	14.70	0.20	(0.66)	4.83	(0.62)	5.45	2008
4.60	4.32	5.34	(0.73)	5.20	12.13	10.11	2.00	(0.31)	14.68	0.22	(0.88)	4.78	(0.79)	5.57	2009
4.64	4.32	5.42	(0.78)	5.13	12.22	10.20	2.04	(0.31)	14.73	0.26	(0.81)	4.81	(0.79)	5.60	2010
4.70	4.39	5.48	(0.78)	5.21	12.26	10.33	2.04	(0.31)	14.78	0.26	(0.82)	4.87	(0.79)	5.66	2011
4.71	4.38	5.56	(0.86)	5.10	12.26	10.34	2.02	(0.33)	14.79	0.26	(0.73)	4.89	(0.86)	5.75	2012
4.78	4.47	5.68	(0.90)	5.21	12.27	10.35	2.02	(0.33)	14.80	0.26	(0.74)	4.96	(0.89)	5.85	2013
4.75	4.43	5.69	(0.94)	5.26	12.31	10.36	2.04	(0.33)	14.82	0.28	(0.83)	4.93	(0.96)	5.89	2014
4.60	4.46	5.68	(1.08)	5.26	12.32	10.39	2.08	(0.33)	14.79	0.32	(0.80)	4.85	(1.04)	5.89	2015
4.62	4.48	5.80	(1.18)	5.36	12.36	10.39	2.10	(0.33)	14.80	0.34	(0.88)	4.86	(1.13)	6.00	2016
4.67	4.51	5.91	(1.24)	5.43	12.42	10.46	2.12	(0.33)	14.83	0.36	(0.93)	4.91	(1.18)	6.10	2017
4.80	4.51	5.96	(1.17)	5.47	12.43	10.52	2.11	(0.34)	14.85	0.36	(0.96)	4.99	(1.16)	6.15	2018
4.85	4.42	5.87	(1.02)	5.45	12.45	10.53	2.11	(0.34)	14.86	0.36	(1.02)	4.99	(1.09)	6.08	2019
5.03	4.53	5.99	(0.96)	5.52	12.43	10.52	2.11	(0.34)	14.85	0.36	(1.00)	5.15	(1.04)	6.19	2020
5.19	4.64	6.08	(0.89)	5.66	12.50	10.57	2.10	(0.35)	14.91	0.36	(1.02)	5.30	(0.98)	6.28	2021
5.03	4.69	6.01	(0.98)	5.56	12.52	10.61	2.11	(0.35)	14.93	0.37	(0.87)	5.20	(1.01)	6.21	2022
5.01	4.67	6.00	(0.98)	5.47	12.56	10.61	2.10	(0.36)	14.94	0.37	(0.81)	5.18	(1.02)	6.20	2023
5.35	4.70	6.03	(0.68)	5.59	12.59	10.63	2.11	(0.36)	14.96	0.38	(0.90)	5.44	(0.81)	6.26	2024

Source: calculation based on appendix 1